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**ADAPTIVE DEEP REINFORCEMENT LEARNING FOR DYNAMIC FLOOD-
AWARE EVACUATION ROUTING IN URBAN AREAS OF MANILA CITY
USING A GIS-BASED APPROACH**

A
Thesis

Presented to the Faculty of Computer Science
College of Computing and Information Technologies
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Specialization in Machine Learning

BY

John Angelo A. Basilio

Iber Joseph P. Bonilla

Robbie Christian Emmanuel E. Espaldon

Andrew Laurence T. Fat

Dhan Micheal L. Tamparong

Ms. Marizkays P. Jamison
Thesis Adviser

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ABSTRACT

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CHAPTER 1

INTRODUCTION

The Philippines is an archipelagic country in Southeast Asia [1] that is highly exposed to hydrometeorological hazards, making flood risk reduction and resilient urban planning a major societal priority. Every year, about 19 to 20 tropical cyclones approach the Philippine Area of Responsibility (PAR) [2] and about 8 to 9 of these storms make landfall, bringing heavy rain and flooding to cities that threaten lives, infrastructure, and businesses. The population of the City of Manila is approximately 1.9 million per the 2024 census [3] and affects many important assets and services, which makes the social and economic impacts of urban floods even worse. Historical events such as Tropical Storm Ondoy demonstrate how rapid urban inundation can produce severe humanitarian and economic impacts across Metro Manila and adjoining regions [4].

During severe flooding, government and rescue agencies deploy specialized vehicles and organized evacuation efforts to move people to safety [5]. However, many residents must still attempt self-evacuation using private vehicles or on foot. Existing navigation tools and commonly used re-routing strategies do not explicitly account for dynamic flood conditions, which can lead road users into routes that appear passable but are in fact hazardous. This gap motivates the development of routing systems that are specifically designed to be aware of floods and can adjust to quickly changing conditions.

The problem now is that there is no navigation companion that can lead people to safety while also avoiding floods and trying to go through places that may seem passable but are actually not, can put one's life in more risk. Which is why the researchers of this paper sought to figure out a way to make navigating through flooded streets safer with the use of Deep Reinforcement Learning (DRL) and Geographic Information System (GIS) data.

There are routing techniques used at times of flooding in the context of the Philippines—algorithms, crowd sourcing applications, and literally looking at the flooded roads before deciding to leave; these methods are deemed ineffective compared to DRL agents and from here on will be classified as "traditional methods" for re-routing.

This study mainly contributes to Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities [6] by enhancing urban resilience to flood risks. The proposed system integrates GIS-based flood modeling with adaptive deep reinforcement learning for real-time evacuation routing, aiming to mitigate exposure and harm to urban populations during flood incidents events by directly contributing to objectives such as decreasing disaster-related deaths and the number of people affected by water-related disasters (target 11.5) and improving urban planning and management through data-driven tools (target 11.b) [6].

The researchers will develop operational strategies for flood-aware routing that may be included into municipal emergency response and urban mobility systems, helping cities to respond faster, route evacuees more safely, and make more informed, spatially explicit decisions during floods.

Geographic Information System (GIS) and spatial analysis have become known as essential technologies for the representation, analysis, and visualization of flood hazards and their effects on urban transport networks within the comprehensive field of disaster risk management. [7] [8]. GIS provides the conceptual and technical framework necessary to connect locations to multiple data layers such as elevation, land cover, road networks, flood extents, and critical facilities [9]. This system enables the spatially explicit assessment and visualization used by planners and responders. City flood planning often uses GIS flood maps and hydraulic models to predict how far and how deep water will spread, identify vulnerable sites, and help plan evacuations. However, many routing systems used by the public and authorities don't take real-time flood forecasts into account, so they can offer inaccurate information about whether roads are passable during an event [10].

Reinforcement Learning (RL) provides a formal framework for sequential decision making under uncertainty, usually modeled as a Markov Decision Process (MDP) in which an agent observes states, selects actions, and receives rewards that encode desirable objectives while learning from interaction with an environment [11]. Deep Reinforcement Learning (DRL) combines reinforcement learning with deep neural networks and has been used to solve routing and evacuation problems. Simulation studies show it can improve adaptive behavior, coordination among multiple agents, and the robustness of learned policies [12]-[14].

Soft Actor-Critic (SAC) is a sample-efficient, off-policy DRL algorithm that adds an entropy term to expected-return maximization to promote exploration and stable

learning, making it well-suited for continuous or high-dimensional control and routing under uncertainty [15]. Developing flood-aware routing as an MDP and training SAC agents in spatially transparent, GIS-integrated simulators provides a method to develop policies that balance travel time and safety amid uncertain and temporally fluctuating flood conditions.

The researchers will put a reinforcement learning-based routing system into action in actual cities that requires a realistic sensor and data pipeline. IoT prototypes for monitoring water levels and detecting floods frequently use affordable ultrasonic sensors like the HC-SR04, providing near real-time distance estimations when calibrated properly. The Firebase Realtime Database can get these measurements so that model inputs and visualizations can happen in real time [16] [17].

To address this gap, this study proposes an adaptive, flood-aware evacuation routing system that integrates Geographic Information System (GIS) data with deep reinforcement learning (DRL). The researchers formalize the routing problem as a Markov Decision Process (MDP) where states encode spatial context and flood-related observations, actions represent routing selection, direction or speed control, and the reward structure balances safety, travel time, and robustness.

The researchers will use the Soft Actor-Critic (SAC) algorithm, a state-of-the-art off-policy deep reinforcement learning (DRL) method for control. SAC is well suited to continuous or high-dimensional action spaces and improves learning stability through entropy regularization.

To adapt to real, time-varying conditions, the system deploys low-cost HC-SR04 ultrasonic sensors at selected areas in Manila City to collect near real-time water level and obstacle data. Sensor readings stream to an operational Firebase Realtime Database [18], which feeds the DRL agent and the GIS visualization layer. A feasible decision-support path for evacuees, first responders, and local government units is established by integrating sensor data with Quantum Geographic Information System (QGIS) visualization and a mobile-compatible web interface [10].

To demonstrate operational feasibility, the proposed system implements a hardware-in-the-loop pipeline that connects low-cost edge sensing to the DRL agent and GIS visualization. Ultrasonic distance sensors mounted at key locations provide local water-levels that are read by ESP32 microcontrollers, packaged as JSON records that contain distance, location name, latitude and longitude, and transmitted to a

Firestore Realtime Database for near-real-time data ingestion. The SAC, together with the MDP agent, uses both GIS features from the QGIS environment and real-time sensor telemetry to generate flood-aware routing decisions, which are visualized on an OpenStreetMap (OSM) and delivered to users via a mobile-compatible web interface that is created using Leaflet or TomTom. This end-to-end design integrates field sensing, learning, and spatial delivery into a replicable pipeline that supports both simulation-based training and prototype field validation.

1.1 BACKGROUND OF THE STUDY

The rapid urbanization, complex drainage networks, and high population density have increased the exposure and vulnerability of Philippine cities to flooding, while changing rainfall patterns and the country's frequent tropical cyclones intensify these risks. National and international assessments repeatedly identify the Philippines as one of the countries most at risk from disasters, emphasizing the value of proactive risk reduction and operationally relevant tools that support timely decision-making during sudden incidents [19] [20]. Recent climate summaries and hazard reports further illustrate annual fluctuations in cyclone activity and intense precipitation events that contribute to urban flood risks, emphasizing the necessity for adaptive, data-driven evacuation and routing systems [21].

Flood-related disasters are one of the most important issues for urban emergency management, especially in densely populated Philippine cities where heavy rain and tropical cyclones often cause quick flooding and infrastructure failures. The Philippines averages 19-20 tropical cyclones entering the Philippine Area of Responsibility (PAR) each year, with about 8-9 making landfall. These storms bring heavy rain and flooding that threaten lives, livelihoods, and economic assets [2] [22].

Building on this point, existing statistics and publications [2] [4] indicate that these recurring events underscore the necessity for the development of evacuation planning systems that are more responsive and adaptive to changing conditions. Storms and major floods in the past have done a lot of damage to the region's economy and infrastructure. Particularly, Typhoon Ulysses caused heavy urban flooding in 2020, which resulted in extensive flood damage, with Manila City seeing its worst flooding in years [23].

Traditional emergency evacuation routing methods are mostly static, relying on pre-established routes or offline optimization that cannot adjust in real time to quickly changing flood extents, road closures, or localized hazards during an event. Similarly, numerous regional traffic and transport models are developed for planning or offline simulation, lacking the temporal resolution and behavioral detail required for real-time emergency decision-making [24]. These constraints reduce situational awareness and may direct evacuees onto hazardous or obstructed pathways during rapid flood occurrences—an operational deficiency this research seeks to address.

Structural drainage improvements, floodways, and other physical interventions remain essential, but they often require substantial time and investment to implement. Meanwhile, floods might suddenly block roads and cause damage hidden beneath the floodwater, making pathways impassable for people. While staying at home makes sense at times of flooding, there are people who choose to evacuate on their own while they can, as their current situations can become worse and make evacuation at a later time impossible to do.

Crossing the flooded streets a resident formerly recognized can be very challenging where hazard increases including infections, unknown objects in the water, and the overall uncertainty associated with navigating unclear waters. The phrase “Turn Around, Don’t Drown” [26] serves as a precautionary saying for anyone attempting self-evacuation, as traveling without a visible route can be more harmful than beneficial. The problem is that prior knowledge of the area is a must, excluding people who are unexpectedly stranded or those who struggle with navigation.

Routing during flooding using deep reinforcement learning has not been explored to the best of the researcher’s knowledge, but there are studies that use machine learning and algorithms that achieve similar goals like navigation, routing, and adaptivity. The problem with that approach, however, is that it does not consider possible sudden changes or miscalculations. It would be a reactive gesture to recalculate when problems occur, which in some situations especially on times that the user is evacuating, can be difficult. The main advantage of using DRL for the same job is that it can understand the route better, possibly taking a slightly longer route because it understands that the shortest path is actually riskier based on how it was trained. This approach now becomes proactive and makes it better by avoiding the situation that requires the user to deal with decision making while in the process of evacuation.

This study contributes an integrated GIS-based simulation and deployment prototype that combines dynamic flood modeling, live low-cost sensing, and SAC-based DRL routing for adaptive, flood-aware evacuation. The researchers will create a spatially precise environment that connects road networks with dynamic flood and passability models, develop and test SAC-MDP routing agents across flood scenarios, and demonstrate an end-to-end prototype that uses HC-SR04 sensor data from Firebase in the learning and visualization pipeline to provide route guidance through a mobile web app backed by QGIS.

The beneficiaries of this study involve citizens and commuters in Manila City, whether they are vehicle or pedestrian evacuees, local government units, emergency responders in barangays, urban planners, and developers of decision-support systems. The expected practical results include better success rates for evacuations and less time spent on dangerous road segments during floods, as well as operational decision support for emergency management, a dataset, and a simulation framework for future research.

The data pipeline that was built consists of a lightweight IoT device, which connects to a cloud or database, then to a DRL agent, followed by GIS, and lastly to a web application. HC-SR04 ultrasonic sensors are used to determine the distance between water and obstacles. ESP32 microcontrollers read the sensors and transmit JSON packets to a Firebase Realtime Database, which contains the distance (cm), location (string), and coordinates (latitude, longitude). This ingestion pipeline is currently operational in a web application that utilizes JavaScript to authenticate with Firebase, retrieve sensor data, and update or generate map markers and popups on a Leaflet map layer. The QGIS environment provides additional overlays that include flood layers, traffic, and evacuation centers so that users can see a combined view of the situation. During the simulation and training phase, the SAC agent learns in a GIS-based QGIS environment that shows the MDP the geographical state. In deployment, the trained policy consumes live sensor observations from Firebase to adapt routing recommendations in near real time. This pipeline allows this study to assess both entirely simulated policy behavior and limited real-world adaptability using low-cost sensing.

1.2 STATEMENT OF THE PROBLEM

Rapidly changing urban floods in Manila City produce highly dynamic and spatially diverse hazards that frequently change road passability within short time frames and put commuters, residents, and visitors at significant risk. Current public navigation systems and traditional evacuation-routing methods like shortest-path routing, static pre-planned routes, or offline optimization fail to consider time-varying flood dynamics, localized obstructions, or incomplete and inaccurate sensor data. Thus, these methods may direct evacuees into hazardous or inaccessible streets, leading to delayed rescues, injuries, and avoidable social and economic losses. Considering the Philippines' high annual cyclone activity and Manila City's dense population and asset concentration, failing to provide adaptive, flood-aware routing and decision support during sudden floods will prolong human vulnerability and increase operational strain on emergency responders.

The operational decision support for flood evacuation is further hindered by limiting integration among real-time, cost-effective distributed water-level sensors, spatial flood models, and adaptive decision-making methods that can function under partial observability and unreliable telemetry. In the absence of a strategy that incorporates near real-time observations, evaluates changing capabilities of being navigable, and prioritizes evacuee safety amid uncertainty, municipal responders and the public will persistently encounter avoidable delays, misrouting, and elevated risk during rapid urban flood events, especially for vulnerable populations, commuters, and visitors unfamiliar with local hazards.

1.3 OBJECTIVES OF THE STUDY

1.3.1 General Objective

This study, entitled “Adaptive Deep Reinforcement Learning for Dynamic Flood-Aware Evacuation Routing in Urban Areas of Manila City using a GIS-based Approach,” aims to create and assess an integrated, adaptive evacuation-routing framework that enhances evacuee safety in dynamic urban-flood conditions by integrating GIS-based spatial simulation, real-time sensor data, and reinforcement learning-driven decision-making. The duration of this study will take place from August 2025 to February 2026, with task allocation determined by the researchers' discretion.

1.3.2 Specific Objectives

- To create a spatially explicit simulation environment that integrates road networks, Digital Elevation Models (DEMs), and dynamic flood propagation or passability models, making it ideal for training and testing the DRL agent.
- To instrument, calibrate, and validate HC-SR04 ultrasonic sensors at selected monitoring points and send processed water-level in real time to a Firebase Realtime Database for utilization by the learning and visualization pipeline.
- To formalize the evacuation routing problem as a Markov Decision Process (MDP) by specifying states, actions, observations, and a reward function balancing safety, travel time, and robustness.
- To train and evaluate Soft Actor-Critic (SAC) agents in the GIS-based simulator and compare their performance against representative baselines using measurable metrics—evacuation success rate, time-to-safety, proportion of flooded segments navigated, and adaptability across scenario variations by the conclusion of the experimental phase.
- To implement an end-to-end QGIS-based prototype that ingests live HC-SR04 sensor data through ESP32 to Firebase, executes the SAC agent trained within the QGIS-integrated environment, and visualizes recommended flood-aware routes and contextual flood layers in QGIS as well as a mobile-compatible web application for evacuees and operators.
- To evaluate system performance by measuring operational metrics such as safety, latency, and reliability, as well as durability against sensor noise and partial observability, and to ensure user-centered accuracy in route guidance. Subsequently, do a statistical analysis of the results, identify failure modes, and formulate suggestions for system training and possible operational deployment.

1.4 SIGNIFICANCE OF THE STUDY

Manila City experiences flooding regularly, severely threatens large groups of people and important infrastructure. This creates an urgent need for operational tools that facilitate quicker decision-making and protect evacuees during the rapid events associated with heavy rainfall, which demonstrate the effects of urban flooding on both a large scale and on people. This study addresses the need to develop a complete sensor-driven, GIS-based decision support pipeline that combines dynamic models for flood propagation and road passability, real-time HC-SR04 telemetry through Firebase, and SAC deep reinforcement learning to create suggestions for safe, flood-aware evacuation routes. The study aims to help residents and commuters, local government units (LGU) and first responders, and future researchers by integrating spatial analysis, low-cost sensing, and learning-based routing. These findings will fill in an important methodological and applied gap in urban flood response.

1.4.1 Residents

The proposed system provides evacuees with context-aware, flood-aware route recommendations that reduce the likelihood of encountering impassable or hazardous streets, helping to minimize injury, evacuation delays, and time spent exposed to flood hazards. By combining GIS-based flood layers, near real-time HC-SR04 sensor telemetry, and SAC-trained routing policies, the system can favor slightly longer but safer routes learned from experience rather than shortest-path directions that may be unsafe during floods—thereby improving evacuation success and timeliness for both motorists and pedestrians in Manila City.

1.4.2 Local Government Unit

Local Government Unit (LGU) benefit from this study and project because of the value of understanding how to find the ideal way for rescue teams to navigate the affected road networks of the area. With the help of this study, LGUs are opened up for another way of thinking when it comes to mapping out possible temporary shelters and also creating better preventive solutions to mitigate commonly flooded areas. Effectively providing a new way of thinking about evacuation planning and infrastructure improvement.

1.4.3 Future Researchers

This study provides a simulation environment that is spatially explicit, a hardware-in-the-loop dataset, and reliable evaluation algorithms that address a methodological gap at the convergence of dynamic flood propagation, high-resolution GIS, low-cost sensing, and DRL-based routing. These artifacts and documented experiments will support subsequent research, allow independent validation and enhancement of the approaches, and reduce the barriers for real pilots or integration with municipal systems.

1.5 SCOPE AND DELIMITATION

1.5.1 Scope of the Study

This study, “Adaptive Deep Reinforcement Learning for Dynamic Flood-Aware Evacuation Routing in Urban Areas of Manila City using a GIS-based Approach,” focuses on developing, implementing, and evaluating an integrated, sensor-informed decision-support pipeline for adaptive evacuation routing under dynamic urban flood conditions.

The researchers chose to use a Deep Reinforcement Learning (DRL) agent and train it under the QGIS environment approach, with a geographic focus on the City of Manila. This study is primarily simulation-based, where the researchers will build a replicated road network of Manila City and a spatial environment that combines road geometry, Digital Elevation Models (DEMs), and simplified flood propagation and passability models—evaluating the routes available to reach the nearest evacuation center and dynamically reroute to avoid flooded road segments with the help of what is has learned from the risk levels associated with the varying flood depths.

This study aims to implement an end-to-end prototype that integrates the trained agent with a QGIS-backed spatial environment and exposes flood-aware route recommendations through a mobile-compatible web interface. The study started on August 2025 and is expected to be finished at the end of the 2nd semester of the College of Computing and Information Technology (CCIT) Department, February 2026.

1.5.2 Delimitations of the Study

This study is limited to selected areas within Manila City and uses a simulated road network rather than full citywide live operation to keep the project focused and feasible. The study uses HC-SR04 ultrasonic sensors that are mounted and calibrated solely for demonstration and integration testing, which means that it will be lab tested rather than a full roadside deployment and long-term field installation. This is to ensure that all efforts are focused onto the development of the agent.

The flood modeling of this study will use simplified, routing-focused representations of inundation and passability rather than high-fidelity hydraulic simulations. This study uses the Soft Actor-Critic (SAC) as its main learning algorithm, defining the issue as a Markov Decision Process (MDP). The researchers may discuss or briefly evaluate additional reinforcement learning algorithms or multi-agent coordination approaches, but a comprehensive comparison of all deep reinforcement learning methods is omitted.

The prototype is a proof-of-concept with limited field validation and controlled tests that doesn't include a full operational rollout, long-term maintenance, or institutional deployment with LGU partners during the study period. This study does not include the creation of a fully offline or local deployment, as the web application prototype requires an internet connection for normal operation.

This study focuses on route selection and passability during flood scenarios without performing detailed microscopic traffic simulation throughout a complete urban transport network. Traffic interactions will be estimated or regarded as uncontrolled where necessary.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 RELATED LITERATURES

2.1.1 Flood Risk and Urban Resilience

Flooding continues to represent a significant threat to cities like Manila as it can lead to disruptions in transportation systems of all types, threats to security and resiliency that increase vulnerabilities in cities. Flooding is represented in the Manila Disaster Risk Reduction and Management Plan 2021 to 2028 (City Government of Manila, 2021) as of a regular nature concerning cities whereby adapted routing systems are necessary for evacuations. Recent literature has shown that there is a growing recognition by the city planners to make use of GIS based systems for the management of disasters in the city as an effective method of allowing a greater spatial awareness about risks from flooding in cities is seen as necessary. It assists the decision making processes for public officials and people who live in the environment concerned about these issues and serve as a source of information and planning (Serrano, et al., 2024; Canillo & Hernandez, 2021). There is p38 however a growing emphasis by these studies upon visualization and prediction of flooding and little about adaptation of routing systems to flooding in real-time from the emergency response model as considered here. This area of disaster response, in connection with flooding is one that requires further exploration in connection with the real-time adaptive evacuation systems from live flooding stressors.

2.1.2 Geographic Information Systems (GIS) in Flood Management

Geographical Information Systems (GIS) have evolved from simple passive hazard maps into multi-hazard risk assessment systems. In the flood-prone cities of Asia, GIS assessment of condition can contribute to increased resilience by allowing the most risk prone areas to be identified and exploited for resource allocation by decision makers (He et al, 2024; An et al, 2025). In Manila for example GIS hazard maps are helpful in optimal route planning in evacuations, as height, water availability, floods and routes of access are assessed (Serrano et al, 2024; Canillo & Hernandez, 2021; City Government of Manila, 2021). Furthermore GIS modelling can assist in real time globally appropriate solutions for disaster response (Rezvani et al, 2024; Alabbad et al, 2024). However the same deficiencies of speed of processing of the varied parameters and communication systems that often haunt any accurate estimation of

hazard assessment arise, especially in developing cities. Thus it is suggested by researchers that an integration of IoT and GIS be made, for the processes of hydrology to be sensed in situ, and in real time if necessary, and routes adapted to suit general access (Misra et al, 2025; Bernadeth et al, 2023).

2.1.3 Role of IoT and Sensor-Based Monitoring

The enhancements in disaster management due to IoT (Internet of Things) capabilities for near-real time remote monitoring of networks of sensor systems. This improvement in information flow is enabled by information available from those sensors built into the IoT, with possible sensors that can examine water levels and road conditions, to give information which can be used which can be dynamically updated and used to convert to real-time GIS (Geographical Information System) and AI enabled (Artificial Intelligence) routing modes of transport to be used in response to floods (Mitropoulos et al. 2025; Misra et al. 2025; Bernadeth et al.) 2023 : The IoT networks dramatically enhance situational awareness of the information flow going to emergency response personnel to fashion their responses in the speediest and most effective possible (Arante et al. 2025). The drawbacks in this area are the restricted nature of supplies of energy, the instability of data buildings and the limited scalability of the systems particularly in the area of urban sprawls with vast populations. The problems here exist due to the fact that work is still not being done in the area of the development of fault tolerant networks of the IoT infrastructures which could guarantee that they can provide the necessary reliable means of the quality of the data flows, thus leading to the possibility of data-driven decision making in emergencies.

2.1.4 Deep Reinforcement Learning (DRL) in Evacuation Routing

Deep reinforcement learning (DRL) has surfaced as an appealing possibility for building adaptive evacuation-routing possibilities to mitigate flooding due to their dynamic nature. The learning agents in DRL can adaptively modify their routing choices due to the capability to learn from the experiences obtained in the environment in an actual time basis (Zhou et al. 2024, He et al. 2024). This research shows also that DRL is developed in order to improve its possibilities in coping with gridlocked environments and relationships in routing methods in complex urban environments (Zhou et al. 2025, Takabatake et al. 2025). Multi-agent DRL frameworks for cooperative decision choices in emergencies and evacuation routing have also been bibliographically investigated (Cui et al. 2025, Li et al. 2023). Although these advances have been achieved, the usages of DRL have not hitherto mainly it has proven its use

in traffic or airport environments but have no examples hitherto reverted to urban floods. Hitherto it is wrong to say that DRL has given examples of being trained in the hydrological and spatial data in the urban context of Manila for evacuating purposes because of flooding.

2.1.5 Integrating GIS and DRL for Flood-Aware Routing

The synergy of GIS and DRL provides an effective matrix for adaptive routing for flood evacuation. Research indicates that coupling the spatial factors of GIS with DRL networks can yield dynamically optimal routing as flood conditions fluctuate (Alabbad et al. 2024; Panfilova et al. 2024). This is particularly useful in urban environments such as Manila, where large populations and rapidly changing flood conditions demand ongoing decision making in real time. Other research has begun to combine models of hydrodynamics with DRL to improve the safety and accuracy of flood evacuations (Zhou et al. 2024; Cui et al. 2025; Gomes et al. 2024). However, much of this research is still in the simulation capacity without extensive validation in the real world. Finding a means whereby the spatial dimensioning capacities of GIS can be coupled with the adaptability of DRL in an ongoing disaster capacity remains a primary challenge.

2.1.6 Smart Cities, Disaster Preparedness, and Sustainability

Smart city frameworks allow for an integrative approach to the implementation of geographic information systems alongside IoT and deep reinforcement learning within disaster management systems. Furthermore, an intelligent sharing of data, real-time monitoring, predictive analytics and improved disaster preparedness are achievable via such frameworks, according to He et al. (2024) and Arante et al. (2025). In addition, it is also important to note that sustainable development frameworks may consider flood-aware routing systems one of the most valuable streams of research for urban resilience (Panfilova et al., 2024; Takabatake et al., 2025). Still, published smart city research appears to have focused in the main upon research for the developed world with little exploration of their applicability to the developing world including in that area of urban resilience, essentially Manila, where issues of infrastructure reliability and accurate data still remain as challenges (Betterle & Salamon, 2024; Gomes et al., 2024).

2.2 RELATED STUDIES

Quantifying Flood Hazard and Risk

Understanding flood hazards and risks is also understanding that static and binary classifications are not enough when it comes to creating solutions. Multi-factor approaches are the way to go when it comes to understanding how floods happen and what can happen because of them. A study in the urban areas of Bangalore found in [50]. found that for their country, there are main key reasons for urban flooding are land use mismanagement, community disinvolvement, and the disappearance of natural water ways. According to the study, one of the main ways to prevent flooding from happening is to have better ways to let the water flow to avoid it from filling the streets. Solutions for this include better flood control infrastructure, incorporating more green space, and also handling the land use management of areas. This is where this research paper comes in, providing a temporary solution for a problem that will be fixed after the time when the root cause is actually handled properly.

When quantified, flood risks become easier to measure and visualize, and the risk becomes calculatable and effects predictable. This is why the researchers of this paper looked at other studies to fully understand risk and how they can scientifically refer to the effects of floods. As found in [51], adjusting the quantitative scoring metric based on the application of it while also having a multi-factor approach in order to make the scoring fully comprehensive and able to score appropriately for the use case is highly recommended. In order to understand how to adjust the risk scoring metrics for this paper, the researchers looked at [52]. This paper looked at the factors of how to quantify the risk of floods and also gives the equivalent of risk score to how it can be interpreted. Defining both as the basis of this paper's own risk scoring system and how the researchers can understand the significance of the scoring method. Both metrics made by Maranzoni and Jafar are good baseline for understanding risk, but the former lacked the latter's breakdown on the meaning of the scoring system. This study will build upon the methodology of [53] by integrating unique factors relevant to the local cultural context, which were not present in the cited works.

While solutions against floods are mostly physical solutions like infrastructure building and adjustments, those will take time to build. Understanding what solutions have been executed provides a foundational context for this study. Talking about flooding in urban areas, it suffers greatly from floods if the area is not planned for correctly or corners were cut in the making of flood control projects. As demonstrated

in [53], flooding in urban areas not only poses hazardous risk for people, but also for the economy. Not only is land-use a factor in how inundation should be handled, but the socio-economic factor of it should also be integrated as well. The researchers in [54] investigated incorporating the input of the local residents of Manayunk and Eastwick, Philadelphia regarding how floods occur in their area. While this research paper will not be utilizing crowdsourced data, as it will be focused on the quantitative methodology aspect of assessing the problem, incorporating qualitative data in the future will help with understanding a deeper context in the minute details when it comes to how floods behave in an area can significantly help the DRL model in deciding the best route to take.

Review of Flood Modeling and Evacuation Strategies

The researchers in [55] explored the use of deep learning models in predicting possible flood levels in Hong Kong with the use of spatiotemporal data and hydrological models for simulating. With the aim of improving early warning measures for their country, they succeeded in their original plan. However, this approach of utilizing digital terrain models of a city is not applicable to all cities, and especially the Philippines. The solution for this is with the use of physical sensors which this paper will be exploring. This effectively makes the exploration of understanding where floods are to be more accessible to more countries as sensors are faster to produce than mapping big cities digitally. Conversely, creating better infrastructure is one of the other ways cities can become more prepared should excessive rainfall occur. The methodology of [56] developed a study establishing a nonlinear optimization framework using Sequential Least Squares Quadratic Programming (SLSQP) in combination with Geographic Information System (GIS) spatial data to determine the best storm-drain pipe dimensions and mitigate urban flooding episodes. Their method efficiently determines segments of pipelines hydraulically insufficient by optimizing the diameter of pipes as the key decision variable. However, their system is primarily long-term, capital-based infrastructure design and often imposes significant funds and time barriers to implementation. Building on the foundation of using GIS to support flood hazard mapping, this thesis shifts the focus from static design of infrastructure to dynamic operational response planning. Instead of optimizing pipes' dimensions, we introduce a concept relying on Adaptive Deep Reinforcement Learning (DRL) using real-time and expected GIS data (e.g., rainfall intensity, inundation mapping, and traffic intensity) to plan optimum, dynamic evacuations routes during flood incidents.

Deep Reinforcement Learning (DRL) for Routing and Navigation

While traditional routing algorithms, like Dijkstra's and A*, exhibit effectiveness in static situations, they become inefficient for dynamic conditions, specifically urban flooding in the case of this study. Which is why researchers are looking for better ways to develop more robust adaptive navigation algorithms. Deep Reinforcement Learning (DRL) algorithms address the limitations that traditional routing algorithms face, but with a caveat: DRL algorithms require more data and a good understanding in optimization of its learning process in order to fully capitalize on its adaptive abilities. The authors of [57] [58] looked into exploring the practical use of DRL in autonomous navigation systems. In their study they found that DRL algorithms are great for this application, but looking into key parameter tuning is paramount into differentiating a good vs great performance. These studies give foundation to the exploration of utilizing DRL algorithms for real-time and complicated environments as opposed to traditional algorithms that fall short behind the former's decision-making capabilities.

As studies regarding DRL algorithms applied in routing and navigation continue to be explored, there are some notable ones that highlight the use of optimization techniques for more efficient training and choosing the right algorithm for the job. The methodology of [59] [60] are both key studies in this regard. The former utilizes Double DEEP Q-Network and Deep Deterministic Policy Gradient (DDPG) algorithms, and the latter uses Deep Q-Network (DQN), Proximal Policy Optimization (PPO), and Trust Region Policy Optimization (TRPO). Training optimization is one of the most important steps to achieve high success rates in navigational tasks for agents. Applying this logic and a formulated metric, the algorithms' performance (DQN at 75.2%, PPO at 82.6%, and TRPO at 78.1%) demonstrated the importance of carefully tuned scoring metrics which account for parameter tuning, exploration-exploitation balance, and reward shaping [60]. Tabakis on the other hand highlights choosing the right algorithm for the job, they found that DDQN works best in static environments, where adaptability may not be the main concern but rather consistency [59]. While DDPG performs better in dynamic conditions due to its adaptability and continuous action capabilities. Effectively serving as guidance for the practical implementation of the algorithms they utilized in real-world autonomous applications. These are the underlying rationale behind the approach of the researchers in this paper. By combining the findings and recommendations of both papers in order to develop an adaptive algorithm in the application of flooding scenarios.

Application of DRL agents with real-world data is already being explored, from traffic management to robotics. In order to address the challenge of optimizing routing for Autonomous Vehicles (AVs), [61] explored the use of DRL in a simulated urban environment in the city center of Bari, Italy. They found that utilizing a modular architecture with DRL can be beneficial in the context of urban routing for AVs. On a similar note, [62] tackled the problem of enabling mobile robots to navigate and explore independently in unknown environments while only being limited to sensor data. They found experience pool optimization and goal orientation strategy to be the key optimizations for their use case. The former enhances the utilization rate of positive rewards, while the latter prevents blind exploration by robots. While these are foundational studies, they do not address this paper's core problem: integrating optimal algorithm selection and specific tuning for a flood-aware DRL navigation system.

GIS and Environment Modeling

Geographic Information System (GIS) have become a cornerstone of modern flood management, providing a robust framework for hazard mapping and risk analysis and proves to be a powerful tool in all aspects of natural disaster studies. A study in Malaysia [63] assessed the feasibility of using GIS when it comes to floodings. It proves to be helpful in detailed flood risk mapping, a helpful data metric when it comes to creating solutions for natural disasters. Static flood assessment should also not be taken out of the picture as there are applications that it is best used for. With a similar goal, [64] conducted a study that successfully delineated low, moderate, and high-risk zones in the Majalaya District, Indonesia. The ability of GIS to streamline spatial analysis and integrate diverse data proves to be very useful in the context of hazard mapping, being able to function in a dynamic or static manner.

While static flood hazard maps provide valuable long-term context, dynamic mapping is on the rise as the natural capabilities of hardware and software increases. A study in Indonesia combines the benefits of static and dynamic mapping by giving the spatial context, but with the capability to assess the current conditions of the area. In a simulated environment, Taringan has been able to get and process relevant data to help aid decision making of the user in under 30 seconds [65]. GIS in combination with other tools produces exceptional outcomes in flood risk management. [66] and [67] explored the effectiveness of Geospatial Artificial Intelligence (GAI) with remote sensing and cutting edge geo-computational techniques respectively for flood risk

management. Utilizing Machine Learning, historical and real-time data, satellite, aerial images, remote sensing, and LiDAR technologies to identify flood-prone areas and understand the reasons of flooding in the area. These studies prove that there is a need for exploring dynamic GIS-based tools that help with flood and risk mapping.

Despite being a powerful tool, there are drawbacks to simply relying on GIS alone in the context of flooding. Gathering data about the flood—Accurately and timely—is one of the main concerns in creating a dynamic algorithm based solution. To address this flaw, [68] utilized optical, radar, and terrain data as well as the Google Earth engineer to quantify the flood hazards and exposures of an area. When combined, they were able to gather data faster and process said data quickly for semantic segmentation to understand what sizing of a drainage system is best. In another approach by [69], Synthetic aperture radar (SAR) backscatter data was implemented from Sentinel-1 satellites which produced near-real-time flood maps on a global scale, which was possible with local thresholding techniques. To identify flood-prone zones, [70] explored the use of GIS, spatial analysis, and hydrological modeling. While effective, the exploration of using actual sensors is something that they recommend giving further context of the situation. The use of Internet-of-Things (IOTs) is one of the main ways real-time data can be gathered at an acceptable accuracy and rate. [71] combined the use of hardware placed at strategic places and machine learning algorithms were able to produce accurate urban flooding nowcasting and track the urban global- real-time water levels from a small number of monitoring points. While the studies mentioned are great foundational knowledge, it doesn't explore the use of sensors to make accurate marking of floods in a spatial manner. Utilizing on-site hardware is essential to creating timely and adaptive mapping of risk, a crucial instrument in creating adaptive models and agents for data-driven flood evacuation response.

IoT and Sensor-Based Flood Monitoring

The rise of the Internet of Things (IoT) has provided a robust and accessible solution for real-time environmental monitoring, specifically for urban flooding. Creating new ways to create reliable data gatherers with some even offering it at significantly low prices. A lot of studies explore the use of IoT hardware to monitor water levels. The studies [72] [73] utilized ultrasonic sensors to identify and quantify flood depth levels with the help of Arduinos and micro controllers. Data is sent from the sensor to

the web using different methodologies on each paper, AshokKumar's utilizing ThingSpeak Cloud Platform and Darwis [72] with MySQL approach [73]. While both studies found success in their methodology, there are notable issues that need addressing to deploy hardware in a more robust manner.

The findings of [74] indicate that their approach of using ultrasonic sensors combined with LoRaWAN for data sensors offered reliable data gathering transmitting in Brooklyn, New York. A problem that they faced however is when it is a sunny day, they found that sunlight has a probable effect in sensor gathering, an estimate of 1% increase in depth measurement. On the other hand, [75] explored the use of buoys to possibly avoid this issue. Their findings dictate that utilizing buoys can make the error percentage gap lower, but without discarding the idea of not using buoys, they suggest that even without using buoys errors can still be minimized when deployed in the field while also keeping in mind external factors that can affect sensor data gathering. Power management is also a factor to consider, which is addressed by the work in [76], wherein they designed a system that allows their sensors to be compact and rugged, while also keeping the costs low. They highlight that their solution offers independence from existing power and network infrastructure. This approach made it so that data gathered can be hyper-local but still also be used for a city-wide sensor network.

While the studies mentioned provide a strong foundation for the technical feasibility of sensor-based flood monitoring, a significant research gap remains: the integration of real-time, hyper-local data into an adaptive and intelligent routing system. The studies focus more on data gathering, planning, and creating foundational understanding on how IoTs can be used for detecting floods and even extracting depths. This study addresses the gap of connecting hardware to software in the realm of routing while also utilizing DRL algorithms to create flood-aware, data-backed decisions in choosing optimal routes.

Path Planning with DRL in Markov Decision Process Frameworks

Dynamic and risk-aware path planning is not a new concept in the realm of researching. As early as 2014, researchers have already been exploring the use of Constrained Markov Decision Process (CMDP) in risk aware path planning, incorporating additional costs in the constraints that they set [77] Their study focused

on the infinite total cost of MDPs, which transversely means that it is best when the time to complete a task is unknown, but finite. They found that while their initial theory worked, there are limitations to be faced. Hierarchical methods offered a significant speed up in the process but also produced suboptimal solutions. Risk metrics deteriorated performance because of the consideration of their set cases, affecting 6 out of 10 cases in their testing. Deterioration in Macro-cell splitting because of the imbalance in characteristics tuning, affecting the capability of reducing risk on low-level paths. While the methodological approach and conclusion of their work is now dated, the researchers aim to acknowledge its contribution as a foundational exploration of the subject.

More recent studies have leveraged advance Deep Reinforcement Learning (DRL) algorithms to directly address the uncertainties in real-world environments. Building upon the foundational findings, [78] proposed and explored the use of Distributionally Robust Deep Q-learning (DRDQN) in risk aversion in robot path planning. Their findings include formulation of risk-averse path planning finding problem using MDP, creating a novel DRDQN algorithm that performs well under “worst-case” distribution in certain distances, practical Lipschitz constant-based approximation to lower the bound for worst-case expected making the robust optimization feasible, superior performance over standard DRN—Attaining an impressive “93-98%” success rate vs “76-82%” for the standard DQN based on their testing and scoring metrics, and a tunable risk-aversion parameter to tune the desired level of safety. All of their findings contribute to the use case of DRL and DRDQN in the realm of risk aversion algorithms while also solidifying the need for different ways of utilization of said algorithms in dynamic environments where conditions are not fully known.

While algorithms like DRDQN have proven to be effective, the researchers will be utilizing the Soft Actor-Critic (SAC) algorithm because of its inherent ability in handling continuous action spaces. To improve SAC’s performance in handling entropy, [79] tuned their SAC in a way that it is encouraged to explore while also making it more robust to adaptive behaviors. With a different approach, [80] explored the concept of bounded exploration, directly addressing the data inefficiency commonly found in DRL, effectively making the agent explore more while also collecting informative data better. These solutions aim to make SAC more robust, building upon the base algorithm, which are all helpful in exploring its use in navigation of flooded road networks while also keeping in mind the safety of the user.

2.3 SYNTHESIS

The overview of previous studies and literature shows notable progress in applications of Geographic Information Systems (GIS), Deep Reinforcement Learning (DRL), and Internet of Things (IoT) technologies for flood risk management and evacuation routing systems for flooding. The studies illustrate the increasing capacity of combining spatial analysis, intelligent learning, and real time monitoring to improve urban resilience and disaster preparedness. Global applications have shown that response time and decision making is better during inundation. However, while investigations have confirmed that GIS, DRL, and methods like the Soft Actor-Critic (SAC) and Markov Decision Processes (MDP) work well in respective research areas, these approaches work in silos. This reinforces the need for a composite framework that joins these separated technological fields together—specifically DRL with GIS driven flood modeling into a dynamic, real-time, and adaptive evacuation routing solution dealing with flooding in urban settings.

Risk Quantification and Spatial Modeling (GIS and IoT)

Multiple researchers point at GIS as one of the most important tools when it comes to creating necessary spatial context for hazard mapping and risk analysis [42] [63] [64]. To address the dynamic nature of flooding, some papers also explored the use of IoT and sensor-based monitoring for real time data gathering [38] [71] [73], improving not only the timeliness of flood information, but also creating a deeper understanding in the granularity of flood levels. To visualize connection, GIS is able to integrate multiple hyper-local and real-time data—also known as Geospatial Artificial Intelligence (GAI) [66] [67]. With all of that stated, the theoretical connection between the three main components of this paper have backing studies that all point at the answer that it is a feasible feat.

Theoretical Framework (MDP)

Dynamic and risk-aware path planning is universally modeled through the Markov Decision Process (MDP) framework which is explored in [77] [78]. It works as a standard structure that defines the states, actions, and rewards for an autonomous agent. Studies go back as far as 2014, which explored the use of a Constrained MDP (CMDP) [77] that laid the theoretical groundwork for intelligent agents with the primary objective of safe path selection.

Deep Reinforcement Learning for Adaptivity

While traditional pathfinding algorithms like Dijkstra's and A Star (A*) are efficient in static applications, they fail when complexity and stochastic urban flooding is thrown in the mix [59], hence DRL is a natural progression to improve it.

Addressing the Adaptive Decision-Making Gap

As explored by a multitude of researchers, DRL is superior in creating routing strategies, especially in changing and uncertain conditions [14] [57] [58]. This is the key characteristic that is essential in urban flood scenarios where safety is the number one priority. With that in mind, there are studies that explored the engagement of GIS spatial models with DRL, and it showed that DRL routing systems were able to actively learn from real-time conditions and make corrections in its trajectory based on the flood developments [36] [68].

The Justification of SAC

Soft Actor-Critic (SAC) is among the suitable algorithms for this study. Its inherent focus on entropy maximization and capability to handle continuous action spaces [79] encourages it to explore sufficiently without losing its robustness. Bounded exploration, a modern enhancement for SAC, is also found to boost its data efficiency and convergence speed [80], proving the agent competent to learn and adapt quickly—The main requirement for an emergency response system.

The Research Gap and Novel Contribution

While the aforementioned studies prove that the individual important components of this study work in their corresponding use cases, to the best of the researcher's knowledge, there has not been a study that combines. Therefore, the researchers aim to bridge the technological gap by creating a novel, integrated framework. Combining proven tools to develop an adaptive and risk aware DRL agent, capable of making data backed and optimal decisions prioritizing safety for users in flood affected urban Manila.

Nonetheless, current systems are limited in three key areas: the lack of real-time adaptive capability, inadequate localization of the Gulf of Manila's infrastructure, and weak integration of the technologies. Most models continue to be advised from outside the country and do not take into consideration the specific flood dynamics, high population density, and complex road networks of the Philippines, especially of Manila City. Hence, there is a need for an adaptive, flood-intelligent evacuation routing model

that combines GIS spatial analysis, IoT-based real-time monitoring, and DRL for dynamic learning to improve disaster preparedness and urban resilience in the local context.

CHAPTER 3

RESEARCH DESIGN AND METHODOLOGY

This chapter explains the theories, models, and conceptual principles that support your study. This section shows the academic foundation of your system, model, or method.

3.1 RESEARCH DESIGN

3.2 CONCEPTUAL DESIGN

This study combines ideas from policy optimization and inundation-aware pathfinding to apply Soft Actor-Critic (SAC) reinforcement learning to dynamic evacuation planning under flood circumstances. Three linked components make up the conceptual design: a flood-augmented graph representation encoding real-time hazard data into conventional road systems; an SAC-based agent which navigates this changing environment using ongoing policy correction; and a multi-scale assessment framework that validates performance from localized navigation tasks to city-wide evacuation simulations. By combining adaptive exploration with safety-driven restrictions, the framework aims to bridge the gap between theoretical RL advances and actual implementation in time-critical disaster response.

3.2.1 Problem Formulation

The evaluation routing challenge is modeled as a Markov Decision Process (MDP) given the tuple $(S, A, \mathcal{P}, \mathcal{R}, \gamma)$. The state space (S) reflects the agent's present location, target destination, and real time flood risk information while having three navigational actions: left, forward, and right. The connectivity of the road network determines the transition probabilities ($\mathcal{P}$) and the reward function ($\mathcal{R}$) combines safety, effectiveness, and goal-reaching objectives. The discount factor for future incentives is (γ).

$$S_t = [x_{current}, y_{current}, x_{target}, y_{target}, r_{left}, r_{forward}, r_{right}, d_{left}, d_{forward}, d_{right}]$$

Equation 1. The state representation that is structured as $S_t \in \mathbb{R}^{10}$.

In Equation 1, coordinates (x, y) presently sought for and target coordinates. For neighboring edges (left, forward, right), flood risk levels (r); Distances (d) to possible next nodes in every direction. This approach

guarantees that the MDP includes navigational limitations and spatiotemporal flood dangers straight into the decision-making process.

3.3 PROTOTYPE DEVELOPMENT METHODOLOGY

The development of the Flood-Based Rerouting System followed a software-oriented modular design methodology integrating real-time data processing, spatial modeling, and reinforcement learning for adaptive evacuation routing. The prototype combines conventional routing logic with deep reinforcement learning principles, allowing the system to dynamically generate safe and efficient evacuation paths under changing flood conditions. The methodology consists of two complementary components: the system architecture, which defines the overall computational framework, and the algorithmic architecture, which implements intelligent decision-making through the Soft Actor-Critic (SAC) model.

3.3.1 Machine Learning Model Development

The model architecture is hierarchical, with separate functional levels for data processing, environment simulation, agent training, and performance assessment. The Data Integration Layer compiles hydrologic and spatial data to create a dynamic model of the environment at the most fundamental level. This layer combines real-time flood depth measurements, OpenStreetMap data for road network topology extraction, and a Digital Elevation Model (DEM) for terrain-based flood susceptibility analysis. converted using spatial interpolation to capture changing flooding circumstances.

3.3.1.1 Graph Construction

The dynamic flood risk measures added into edge weights extend conventional road network representations using the flood-aware graph. With vertices V denoting spatial nodes, edges E preserving topological connectivity, and weights W encoding multi-criteria navigation constraints, the graph is given by $G = (V, E, W)$.

$$W(u, v) = \alpha * d(u, v) + \beta * f(u, v) + \gamma * r(u, v)$$

Equation 2. Weighted Edge Function

In Equation 2, $f(u, v)$ denotes the flood depth along the linking edge; $d(u, v)$ represents the Euclidean distance between nodes u and v . The

parameter $r(u, v)$ covers extra risk elements including structural risks or flow velocity. The coefficients α , β , and γ are tunable weights respectively defining the relative significance of path length, flood intensity, and hazard exposure. This formulation guarantees that the graph dynamically represents real-time flood situations and static infrastructure limitations, allowing adaptive routing decisions under ambiguity.

3.3.1.2 Algorithms

For the implementation of the Soft Actor-Critic (SAC) in this study, it is modified to solve the flood-aware evacuation routing problem by simultaneously maximizing a stochastic policy and dual value functions under entropy regulation. This approach is chosen for its sample efficiency and resilience to high-dimensional state spaces, essential characteristics considering the dynamic, and spatiotemporal character of flood hazards encoded in. Unlike traditional reinforcement learning algorithms that give top priority to cumulative reward alone, SAC's maximum entropy framework expressly balances exploration and exploitation, therefore guaranteeing adaptive pathfinding even when flooding patterns change. By means of hierarchical curriculum training, the system progressively exposes the agent to growing flood intensity, therefore reflecting real-world evacuation situations whereby basic navigation takes precedence. abilities before confronting challenging hazard avoidance.

3.3.1.3 Policy Optimization

The policy target $J\pi(\theta)$ in Equation 3 maximizes the predicted return while promoting exploration by means of an entropy regularization word. This dual-objective approach guarantees the agent not just successfully reaches the goal but also has enough stochasticity to adjust to unexpected flood variations.

$$J_{\pi}\theta = E_{S_t \sim D, a_t \sim \pi_{\theta}}[Q_{\phi}(s_t, a_t) - \alpha \log \pi_{\theta}(a_t|s_t)]$$

Equation 3. Formula for Policy Optimization

In Equation 3, the first term $Q_{\phi}(s_t, a_t)$ denotes the anticipated cumulative reward for acting at in state s_t ; the second term $-\alpha \log \pi_{\theta}(a_t|s_t)$ scaled by the temperature parameter α , is the entropy bonus. This bonus encourages the policy to prevent early convergence to poor activities, which is especially

important in dynamic settings such flood evacuation where circumstances might suddenly change.

3.3.1.4 Value Function Updates

The value function $J_Q(\phi)$ in Equation 4 minimizes the temporal-difference (TD) error between the target values y_t and the current Q-value estimates. Adjusted for entropy to keep policy objective consistency, the target combines the immediate reward and the anticipated future return.

$$J_Q(\phi) = E_{(s_t, a_t, r_t, s_{t+1}) \sim D} \left[\frac{1}{2} (Q_\phi(s_t, a_t) - y_t)^2 \right]$$

Equation 4. Formula for Value Function

$$y_t = r_t + \gamma E_{a_{t+1} \sim \pi_\theta} [Q_{\phi'}(s_{t+1}, a_{t+1}) - \alpha \log \pi_\theta(a_{t+1} | s_{t+1})]$$

Equation 4.1. Formula for the target y_t

In Equation 4.1, γ is the discount rate, and r_t is the immediate reward, such as progressing toward the goal or penalty for entering high-risk flood zones while assessing the significance of the prizes ahead. Combining the anticipated return with the entropy bonus, the phrase $Q_{\phi'}(s_{t+1}, a_{t+1}) - \alpha \log \pi_\theta(a_{t+1} | s_{t+1})$ defines the soft value of the next state-action combination. By lowering overestimation bias, the use of a target network ϕ' (different from the present network ϕ) stabilizes training.

3.3.1.5 Hierarchical Evaluation Training

The curriculum-based training method gradually increases the complexity of flood situations to ensure robust learning. Balancing the demand for representational capacity with sample efficiency, the SAC agent starts with a 256-unit neural network and a 100K-experience replay buffer. The curriculum phases $\{S_1, S_2, \dots, S_n\}$ bring flood dangers of ever-increasing intensity, so guaranteeing the agent first achieves fundamental navigation skills before facing hazardous inundation situations.

Algorithm 1 Hierarchical Evacuation Training

Require: Flood-aware graph G , curriculum stages $\{S_1, S_2, \dots, S_n\}$
Ensure: Trained SAC agent π_θ^*

- 1: Initialize SAC agent with network size 256, buffer size 100K
- 2: **for** each stage S_i in curriculum **do**
- 3: Set maximum steps T_i and target success rate τ_i
- 4: **for** episode $e = 1$ to N_i **do**
- 5: Generate scenario (s_0, s_{goal}) according to stage distribution
- 6: Reset environment: $s \leftarrow s_0$
- 7: **for** step $t = 1$ to T_i **do**
- 8: Select action: $a \sim \pi_\theta(\cdot|s)$
- 9: Execute action: $s', r, \text{done} \leftarrow \text{env.step}(a)$
- 10: Store transition: $\mathcal{D} \leftarrow \mathcal{D} \cup \{(s, a, r, s')\}$
- 11: Update agent: $\theta, \phi \leftarrow \text{SAC_update}(\mathcal{D})$
- 12: $s \leftarrow s'$
- 13: **if** done **then**
- 14: **break**
- 15: **end if**
- 16: **end for**
- 17: **if** success_rate $> 2 \times \tau_i$ **then**
- 18: **break**
- 19: **end if**
- 20: **end for**
- 21: **return** π_θ

Figure 1. Hierarchical Evaluation Training

In figure 1, the researchers use a curriculum-based Soft Actor-Critic (SAC) training method intended for progressive mastery of flood evacuation situations. To manage the high-dimensional state space ($S_i \in \mathbb{R}^{10}$), the algorithm begins with a 256-unit neural network and a 100K-experience replay buffer. Training goes via staged situations S_i of progressively more complex floods, each with set maximum steps T_i and success thresholds τ_i . Episodes start with randomly produced start-goal pairs (s_0, s_{goal}) reflecting various flood scenarios for every stage. The agent chooses actions at every timestamp using its present policy $\pi_\theta(\cdot|s)$, carries them out in the flood-aware graph environment G , and saves resulting transitions (s, a, r, s') in the replay buffer. Using mini-batches from this buffer, the SAC agent then modifies its policy and value functions, hence maximizing the goals specified in Equations 3-4. Training for each Early stage termination guarantees computational efficiency while maintaining robust learning if performance surpasses $2 \times \tau_i$. The way the algorithm interacts with graph G (Section 3.2.1.1) guarantees rewards reflect both flood risk avoidance as defined by the weighted edge function and

navigational progress (Equation 2, $W(u, v)$). This hierarchical approach lets the agent create flexible evacuation plans over different flood intensities.

3.3.1.6 Progressive Curriculum Design

The training approach uses a hierarchical six-stage curriculum meant to progressively improve the agent's evacuation skills. Starting with Micro Navigation (50 steps, 30% success target), the curriculum emphasizes short-distance precise movement to create fundamental navigation abilities. Following Zone Traversal (100 steps, 20% target) for inter-zone navigation, strategic waypoints (150 steps, 15% target) introduce intermediate goal routing. The later phases have more difficult situations: Multi-Stage Evacuation (200 steps, 10% target) for complex multi-phase routing and Emergency Protocols (300 steps, 5% target) for high-risk. Handling of scenarios results in Final Mastery (400 steps, 3% target) for long-distance evacuation completion. This staged technique ensures the agent builds strong abilities across the entire set by gradually raising both the spatial complexity (step limits) and performance demands (success targets) from fundamental navigation to sophisticated flood situations, a spectrum of evacuation difficulties.

3.3.1.7 Performance Evaluation Framework

The evaluation approach uses a thorough group of criteria to evaluate how reliable and efficient the evacuation path network is. The percentage of agents arriving at appointed shelters, or the evacuation success rate, is the main barometer of overall system performance. Average Evacuation Time assesses the efficacy of created paths by measuring the average time needed for effective evacuations. The Route Safety Index directly measures the system's capacity to prioritize safety in dangerous circumstances by indicating the proportion of routes that effectively avoid hazardous flood areas. Measuring the robustness of the algorithm to dynamic environmental changes, the Adaptability Score assesses performance consistency under several flood situations. Finally, scalability efficiency looks at system performance as the evacuation zone grows, so guaranteeing the solution stays successful at several operational levels. Regarding success, these signs taken together provide a complete assessment of the adaptability, safety, scalability, and effectiveness of the system.

Data Collection and Preprocessing

3.3.1.8 Geographic Scope and Study Area

This study is focused on Manila City, Metro Manila, Philippines. The selection of this city was based on a number of important considerations. First, with around 71,000 people per square kilometer, it displays the greatest urban density in Metro Manila, which is a highly crowded setting where evacuation route efficiency is most important. Second, the location of Manila City in low-lying coastal areas makes it especially exposed to typhoon-induced storm surges and monsoon flooding, therefore generating frequent scenarios where navigation alerts to floods are absolutely necessary. Thirdly, because Manila City has vital government offices, schools, and big evacuation shelters, it is a strategically important area for catastrophe response planning. At last, the area profits from a well-developed PAGASA flood monitoring network and thorough OpenStreetMap road network coverage, therefore guaranteeing data accessibility for model creation and validation. Covering around 42.88 square kilometers of urban topography, the geographic scope of our study region runs from Latitude 14.54°N to 14.66°N and Longitude 120.97°E to 121.05°E.

3.3.1.9 Road Network Data Acquisition and Processing

The OpenStreetMap (OSM) Philippines extract, accessed via the Geofabrik download server (2024 version), provides the road network data. The extracted file offered information on the road network, and the extraction pipeline began with the download of the whole Philippines OSM shapefile bundle (`gis_osm_roads_free_1.shp`), which covered about 1.2 million road sections throughout the country.

Utilizing the `clip_metro_manila.py` script, which loads the Philippines administrative border shapefile and performs a spatial intersection operation with the Metro Manila polygon boundary (EPSG:4326 coordinates), geographic clipping was done. This filtering process produced 187,423 road sections inside the Metro Manila area while preserving road geometry features, including road classification (*fclass*), names, and connectivity data. The `manila_city_extractor.py` script, which pulled roads particularly inside Manila City administrative boundaries, provided further improvement using PyQGIS spatial query functions. The program chosen Road segments crossing the

Manila City polygon were exported to a new shapefile layer keeping all original attributes and topology.

This resulted in a final road segment collection of 23,845 inside Manila City proper. The categorization of road segments according to their 'fclass' attribute showed that tertiary roads comprised 18% of the network and secondary highways (major arterials) 12%. Local roads make up 35%; residential roads account for 25%; service roads total 10%. Using the EPSG:4326 (WGS84 Geographic) coordinate reference system, all data was kept regularly. A processing pipeline guarantees compatibility with elevation models and flood risk data. Coordinates' precision was standardized to 5 decimal places, giving around 1.1-meter spatial accuracy appropriate for urban-scale route planning.

3.3.1.10 Digital Elevation Model (DEM) Processing

Digital elevation data was taken from the 30-meter resolution DEM tiles covering the Philippines, which were acquired from the Shuttle Radar Topography Mission (SRTM). started with the downloading of SRTM tiles spanning the Manila area (specifically tiles N14E120 and N14E121). These tiles were mosaicked utilizing the GDAL merge utility with bilinear Using interpolation to produce a smooth elevation surface, the mosaicked DEM was resampled to 10-meter resolution and trimmed to the Manila City administrative border.

The researchers used cubic convolution for resampling to capture the more delicate elevation gradients necessary for urban flood modeling. From -2 meters to +108 meters, the elevation values mirrored Manila's coastal topography and inland elevation changes. For every road network node located at (x, y), elevations were sampled from the digital elevation model (DEM).

Using GDAL's RasterIO functions, which carried out spatial lookups based on geographic coordinates, nodes were given their respective height z. The processed DEM employing bilinear interpolation was created based on this information. The mean elevation across all 5,142 road nodes was 5.2 meters, with a standard deviation of 8.4 meters and a minimum of -1.8 meters (beneath sea level in coastal regions) and a high of 47.3 meters in inland locations. Slope

calculations were carried out across all neighboring node pairings to measure the gradient of road segments.

$$slope_{i,j} = \frac{|z_j - z_i|}{distance_{i,j}} \times 100\%$$

Equation 5. Formula for slope i, j

In Equation 5, the slope for nodes i and j with elevations z_i and z_j separated by Euclidean distance i, j was calculated as shown above, where the distance was measured in meters. This slope data enhanced the road network graph by allowing terrain-aware routing that takes both horizontal distance and vertical elevation change into account.

3.3.1.11 Flood Generation Algorithm

The researchers have developed a smart flood generation algorithm producing geographically connected flood patterns aligned with realistic flood scenarios during model training and testing historical information. The algorithm employs a dual distribution technique based on beta distributions:

$$flood_risk_{node} = \begin{cases} \beta(4,1) \times scale & \text{if scenario} = \text{high} \\ \beta(2,5) \times scale & \text{if scenario} = \text{moderate} \end{cases}$$

Equation 6. Formula for getting flood risk nodes

In Equation 6, the scale factor ranges from 0.3 to 1.0 to control general flood intensity. For high-severity scenarios that represent typhoon conditions, $\beta(4, 1)$ produces a mean risk of 0.8, with values skewed toward severe flooding. Conversely, using $\beta(2, 5)$, moderate scenarios create a mean risk of 0.29, therefore simulating expected monsoon flooding. This dual-distribution approach ensures three essential attributes: firstly, spatial correlation, wherein adjacent nodes show similar flood risks reflecting actual water flow patterns; secondly, realistic flood clustering in low-lying areas aligned with Manila's geography; and thirdly, diversity training familiarizes the agent with both regular seasonal flooding and severe typhoon occurrences.

3.3.1.12 Multi-level Flood Severity Classification

To overcome the major constraint of binary flood thresholds that cannot differentiate between tolerable shallow flooding and dangerous deep water, a

new four-level flood intensity classification system was created. Official flood hazard maps supplied by the Manila Disaster Risk Reduction and Management Office (MDRRMO) were produced as part of the Prepared for GMMA Project's Preliminary Flood Hazard Assessment and the Risk Analysis Project's flood exceedance probability mapping. These official maps classify flood vulnerability over Manila City into several degrees, from low flood risk areas to high-risk zones with different annual exceedance probabilities (1/200, 1/100, 1/25, etc.). Table 1 shows how our four-level categorization system turns these official danger gradients into a computational structure appropriate for reinforcement learning-based routing.

Table 1. Multi-level Flood Severity Classification

Level	Risk Range	Depth (m)	Traversability
L1 (Passable)	0-0.3	0-0.5	All vehicles safe
L2 (Difficult)	0.3-0.6	0.5-0.9	High-clearance only
L3 (Dangerous)	0.6-0.8	0.9-1.2	Vehicle submersion
L4 (Impassable)	0.8-1.0	1.2-2.0+	Life-threatening

In Table 1, the depth-to-risk mapping was produced by comparing recorded flood depths from the LGU's historical events records with the flood hazard map classifications. The researchers matched L3 classification with the official maps' designations of High Flood Susceptibility and high exceedance probability zones (1/25 to 1/100 year events). Representing situations where flooding of 0.9 meters or more is anticipated (Dangerous), L4 (Impassable) classifications correspond to L2 (Difficult), meaning depths between 0.5 and 0.9 meters where high-clearance automobiles are still able to function. Low-susceptibility areas are mapped to L1 (Passable), where flooding rarely exceeds 0.5 meters and poses little danger to vehicle operations.

This mapping guarantees that the computational severity levels match exactly the authoritative hazard classifications set forth by the Manila Disaster Risk Reduction and Management Office (MDRRMO). Each severity level includes accessibility evaluations for cars, motorcycles, and pedestrians, hence enhancing multi-modal evacuation planning.

For L1 through L4 respectively, the levels correlate with reward penalties of -0.5, -3.0, -10.0, and -100.0. The exponential penalty growth is a sign of real-world risk, as shown in the flood hazard maps: L4 is 10 times

stronger than L3, while L2 is 3.3 times stronger than L2. The flood hazard maps set powerful deterrents against passing through life-threatening circumstances while also allowing the necessary travel through moderate flooding when no substitutes are available. The implied risk hierarchy of the hazard map, with high flood susceptibility zones calling for substantially more avoidance than moderate or low susceptibility regions, defines this penalty structure.

3.3.1.13 Road Network Graph Construction

Using a multi-stage pipeline executed in the graph builder.py script, the road network graph was built. Node creation started from extracting vertices from OSM LineString geometries depicting, with a spatial tolerance of one meter, road segments with matching coordinates combined to collect junction points. The procedure resulted in 5,142 distinct network nodes spread over Manila City following duplication removal based on coordinate closeness. Edge generation is linked next to nodes. Bidirectional travel was realized for two-way roads by forming While OSM oneway tags generated just single directional edges, coupled forward and reverse sides produced paired forward and reverse edges. The graph captured the whole connectivity of Manila's road network with 27,483 directed edges. Each node was enhanced with several properties to enable flood-aware routing. Position attributes included coordinates in WGS84, flood risk values based on the degree indicating the count of incident edges (averaging 10.7 neighbors per node), LGU flood hazard map rasters, and elevation z_v measured in meters above sea level from SRTM data.

$$weight_{ij} = length_{ij} \times (1 + \alpha * r_i + \beta * road_penalty_j)$$

Equation 7. Formula for calculating edge weights

In Equation 7, edge weights were determined using a multi-factor cost. Function combining geometric distance, flood risk, and road type parameters where $length_{ij}$ is the Euclidean distance in meters between nodes i and d , normalized flood risk at node i assigns a weight of 0.5 to flood risk, β weighs road quality consequences at 0.3, and $road_penalty_j$ ranges from 0.1 to 0.5 with greater values allocated to restricted or congested roadways based on OSM width and lane attributes. With 5,142 nodes and 27,483 directed edges, the average node degree of 10.7 reflects well-connected properties characteristic of urban road networks. multiple routing choices in urban

infrastructure. The graph diameter of 142 hops denotes the longest shortest path, matching cross-city journeys of around 15 kilometers. With an average clustering coefficient of 0.031, a mostly grid-like architecture typical of planned urban growth is indicated; a single connected component confirms full navigability across the study area without isolated road segments.

3.3.1.14 Hierarchical Graph Coarsening

Employing the pre-process graph.py script, hierarchical graph coarsening allows multi-scale route planning that balances computational efficiency with routing accuracy. The coarsening resulted in three graph layers with incrementally decreasing complexity. Layer 1 represents the original fine-grained graph with all 5,142 nodes and 27,483 edges, providing top detail for ultimate path correction. Layer 2, the intermediate representation, was built by merging nodes with degree 2, which represent simple road continuation points without branching decisions. Preserving junction form and removing extra waypoints, this consolidation produced 2,784 nodes and 15,392 edges with a 46% reduction in graph size. Using k-means clustering, Layer 3, the abstract layer, applied more aggregation by clustering nodes within a 500-meter spatial radius and then producing super-nodes representing each cluster and linking them depending on first inter-cluster connectivity.

This merger generated 1,285 nodes and 7,841 edges while achieving a 75% decrease from the original and preserving macro-level connection patterns. This hierarchical planning approach uses the multi-layer representation to have the high-level policy run on Layer 3's 1,285 nodes to choose strategic subgoals and macro-routes, then assigning them to the low-level policy that perfects these plans on Layer 1's full 5,142-node graph for turn-by-turn navigation across the city. While preserving the capacity to identify the best micro-routes, this decomposition lowers the effective action space by 75% during high-level decisions; thereby, computer efficiency is obtained without compromising route quality. While keeping route optimality within 5% of an exhaustive search on the whole graph, empirical testing showed hierarchical planning decreased average planning time from 2.3 seconds to 0.7 seconds.

3.3.1.15 Shelter Location Geocoding

From Manila's Disaster Risk Reduction and Management Office (MDRRMO), official evacuation center sites were acquired; this gave a dataset of twenty-seven declared shelters, including public schools, barangay halls, and community centers. Geocoding, which utilized the Nominatim tool and OpenStreetMap data, converted addresses into geographic coordinates. To integrate with the road network graph, each shelter was mapped to its nearest road node using a k-nearest neighbors search with $k=1$, which provided exact latitude-longitude coordinates and the Euclidean distance metric in geographical coordinate space, ensuring accuracy within a specified range. By keeping spatial correctness inside roughly 50 meters of actual facility sites, shelter destinations were reached through the navigable road network.

Feature Engineering

3.2.1.16 State Representation

At every decision step, the reinforcement learning agent sees a 20-dimensional continuous state vector $s_t \in \mathbb{R}^{20}$, precisely designed to give thorough context on navigation and flood conditions. There are three types of characteristics that make up the representation of the state. Six dimensions include the current node ID and target node ID, both normalized by dividing by the maximum node ID to scale values. Euclidean distance to the target is calculated as the straight-line geographic distance in meters, while Manhattan distance represents the graph-theoretic distance measured in hop count, along with the current latitude and longitude coordinates. To guarantee uniform scaling throughout the geographic area, longitude coordinates are scaled to the range $[-1, 1]$. Features of flood risk (8 dimensions) define the nearby and immediate flood environment. At the agent's location, current node flood risk provides the normalized flood value $[0,1]$. Within a two-hop neighborhood, average, maximum, and minimum flood risk values help the agent foresee future events by outlining local flood patterns.

The computation of the flood risk gradient, defined as the difference between the current and past node flood risk, indicates whether the conditions along the path are deteriorating or improving. While flood encounter count

counts the number of markedly flooded nodes (risk > 0.5) visited, cumulative flood exposure keeps a running total of encountered flood risk. In particular, we added a new current severity level attribute denoting the four-level severity classification (1 through 4) of the flood state of the current node. Routing context Six-dimensional characteristics offer decision-making data; available action count shows the number of viable neighbor nodes, hence indicating route choice availability; distance and direction to nearest shelter allows the agent to consider evacuation as an option when flood conditions are serious, with direction shown as cosine similarity between the vector to path circularity indicator counting recent node revisits to deter ineffective circling behavior; and normalization of step count t/T_{max} offers temporal awareness of shelter and vector to target. The episode's progress is measured against a maximum of 50 permissible steps. A fresh maximum severity function monitors the worst flood level recorded during the current episode, promoting adaptive risk assessment. The layered graph structure, which includes Layer 1 (5,142 nodes) for precise navigation and Layer 3 (1,285 nodes) for strategic planning, allows for automatic modifications of features. Features are automatically modified to the appropriate granularity depending on which policy level is active to enable the state representation to work efficiently over several spatial scales.

3.3.1.17 Action Space

At every stage of choice, the agent picks from a limited action space. Reflecting the erratic topology of urban road networks, the action space varies naturally in size, from 2 neighbors at simple road continuations to more than 20 at main intersections. By graph connectivity lookup, available actions are found to stop the policy from picking incorrect transitions; invalid neighbors are hidden throughout policy sampling, reducing their option likelihood to zero. Early training or in previously unknown graph areas, the fallback system deals with instances when the learnt policy generates actions reflecting nonexistent neighbors. The mechanism determines the geometric center of all legitimate neighbors and chooses the neighbor nearest to the originally intended one once an invalid action is selected. This approach maintains the agent's strategic purpose while ensuring legitimate travel within the graph. Fallback action frequency is monitored as a performance measure, indicating that well-trained policies achieve fallback rates under 5%, which denotes satisfactory graph structure learning.

Model Selection and Architecture

3.3.1.18 Algorithm Justification

Given its particular benefits for flood-aware evacuation routing, Soft Actor-Critic (SAC) was chosen as the central reinforcement learning algorithm. By using past experiences kept in a replay buffer, SAC's off-policy learning strategy first enables extremely sample-efficient training on vast road networks, hence lowering the number of episodes required compared to on-policy approaches. Second, the maximum entropy framework promotes discovery of varied evacuation routes rather than prematurely converging to locally ideal paths by explicit entropy regularization in the objective function, hence encouraging exploration. Thirdly, by reducing overestimation bias in value function approximation, the twin Q-network architecture with clipped double Q-learning offers steady convergence for safety-critical routing choices. SAC's continual adaptation ability via off-policy updates finally enables effective learning over different flood situations without the need for costly on-policy rollouts for every fresh flood setup. In preliminary testing, SAC showed better results than other approaches. On-policy restrictions in methods like PPO required 3–5 times more episodes to achieve the same success rates, while value-based approaches like DQN struggled with instability during training and the large discrete action space (up to 20 neighbors per node). In contrast, SAC's combination of stability, sample efficiency, and exploration capabilities made it the best choice for our flood-aware routing problem or for scenarios with poor routes that overlooked shelter evacuation plans.

3.3.1.19 Neural Network Architecture

The actor network (policy π_θ) provides a stochastic policy that maps state to action distributions. The network architecture consists of an input layer that accepts the 20-dimensional state vector $s_t \in \mathbb{R}^{20}$, followed by three fully connected hidden layers with 256 and 128 units, respectively, utilizing ReLU activation functions to introduce nonlinearity. After the initial two dense layers, batch normalization layers are added to stabilize training by normalizing activations; dropout with probability 0.2 follows. The second layer is used to avoid overfitting. The output layer generates two parts: a mean vector $\mu_\theta(s)$ with tanh activation to limit action preferences and a log-standard-deviation vector $\log(\sigma_\theta(s))$ restricted to the range $[-20, 2]$ to guarantee numerical stability

while letting enough exploration variation. Utilizing the same architectures, the twin critic networks ($Q\phi1$, $Q\phi2$) estimate state-action values to carry out clipped double Q-learning, where each critic accepts concatenated state-action input. The input format is $[s_t, a_t] \in \mathbb{R}^{21}$, which consists of 20 state dimensions and one discrete action index represented as a scalar. With ReLU activations, three completely linked layers are 256, 256, and 128 units that manage the input; batch normalization comes after the first two layers. Last layer yields a single Q-value scalar that reveals the projected total reward for behaving a_t in state s_t . The lower of the two critic ratings is employed for policy updates throughout training to avoid overestimation. While both critics are independently updated using Bellman targets, both actor and critic networks use Glorot uniform weight initialization to establish preliminary weights. Variance is scaled according to layer dimensions to ensure stable gradient flow early in training.

3.3.1.20 Hierarchical Extension

To solve the computational difficulties of planning routes across extensive metropolitan networks spread over kilometers, the hierarchical SAC variation uses a two-level decision-making strategy. Operating on the Layer 3 abstract graph with 1,285 nodes, the high-level policy has each abstract node represent a spatial cluster of about 4–5 original road nodes. This policy picks strategic subgoals and establishes macro-level routing decisions, including which key arterial roads to employ and whether to travel toward the destination or divert to a shelter. The low-level policy operates on the Layer 1 detailed graph, which contains all 5,142 nodes, receives subgoals from the high-level policy, and generates precise turn-by-turn navigation decisions to achieve those subgoals while avoiding local flood risks. By 75% during high-level planning (1,285 nodes vs 5,142), this hierarchical decomposition lowers the effective action space, therefore speeding the calculation of strategic paths. The low-level policy's emphasis on short-horizon subgoal attainment (usually 5–10 hops) also helps to solve the credit allocation challenge and quickens learning of local navigational skills. Using the same SAC goal but with different replay buffers, the two policies are trained together: high-level experiences store abstract state transitions and multi-step rewards. Low-level experiences, which are aggregated based on subgoal fulfillment, reflect minute node-to-node transitions and instantaneous incentives. The hierarchical model showed

especially outstanding performance on long-distance routes more than 10 kilometers during evaluation, with success rates 15% greater than those of the flat policy and reducing planning time by 60%.

Training and Validation

3.3.1.21 Training Protocol

Grid search over important parameters was used to methodically optimize SAC hyperparameters, assessing combinations spanning 25 training runs. Table 2 indicates the optimal configuration, balances converge speed, sample efficiency, and final performance quality.

Table 2. SAC Training Hyperparameters

Parameter	Value	Rationale
Episodes	50	Converge on 5K-node graph
Max steps/episode	50	Realistic route lengths
Replay buffer	100,000	Diverse experiences
Batch size	64	Stable gradients
Discount γ	0.99	Long-term success
Target update r	0.005	Soft target updates
Initial temp. α	0.2	Auto-tuned
Eval interval	5 eps	Progress tracking

Using the flood generation algorithm, the training environment randomized flood situations at the beginning of each episode and sampled equally from either high-severity $\beta(4,1)$ or medium-severity $\beta(2,5)$ distributions. To provide instruction on real evacuation scenarios emanating from populated areas, start-target node pairs were drawn equally from high-density urban nodes with more than 10 connections. Episodes finished either when the target node was reached, upon arrival at a shelter, or upon 50 steps, with shortened episodes deemed as failures to promote optimal routing. Every four environmental measures balance learning effectiveness with sample collection. Starting at 0.2 and converging, automated temperature tuning changed the entropy coefficient α across training, reflecting the change from exploration to exploitation, to roughly 0.05 by the last episodes.

3.3.1.22 Validation Strategy

The validation strategy used rigorous train-test splitting and baseline comparisons to evaluate model generalizing and performance benefits. With flood conditions varied across episodes using beta distribution generators, the training set comprised 1,000 randomized start-target pairings taken from the entire Manila City Road network. Specifically, for routes in northern and eastern Manila regions, the test set comprised 200 held-out route pairs chosen from areas not well represented in training data. Five-fold cross-validation was conducted by partitioning the integrated dataset into five subsets, utilizing four to be trained and one for validation to ensure statistical validity, with results presented as mean and standard deviation across the folds. Three baseline algorithms provided performance benchmarks. Dijkstra's algorithm with uniform flood penalties applied a constant cost multiplier of 2.0 to edges with flood risk exceeding 0.5, representing traditional flood-aware routing. By merging Euclidean distance to target with cumulative flood exposure along the predicted trajectory, a search brought flood cost into the heuristic function. SAC, lacking the sophisticated multi-level severity system, had a binary flood threshold that reflected our past approach, employing a basic flooded/not-flooded classification with a set -5.0 penalty above the 0.5 risk.

3.3.1.23 Model Selection

Final model selection prioritized safety, reliability, and convergence stability using a multi-criteria evaluation approach. Ensuring enough routing dependability for actual deployment, the first criterion demanded a minimum 75% evaluation success rate on the held-out test set. The second criterion reduced average flood exposure to below 0.5 mean risk across all route nodes, therefore stopping the agent from regularly crossing hazardous floods even if theoretically successful. By calculating reward variance over the last ten episodes, the third criterion tested training stability and demanded variance below 2.0 to verify convergence instead of oscillatory behavior.

Evaluation Metrics

3.3.1.24 Route Safety Metrics

The flood severity study presents a complete route safety assessment framework based on the four-level severity classification. Severity distribution measures the flood safety profile by computing the proportion of path nodes at each severity level (L1 through L4).

$$SevDist_k = \frac{|\{n \in path : severity(n) = k\}|}{|path|} \times 100\%$$

Equation 8. Formula for flood severity distribution score

In Equation 8, the k is the level of severity, and $|path|$ is the route length in the nodes among $\{1, 2, 3, 4\}$. This criterion offers understandable knowledge regarding route composition. Routes with any $SevDist4 > 0\%$ are automatically designated as inaccessible, needing urgent replanning, while those with $SevDist1 > 80\%$ are thought of as generally safe and need only little attention. Maximum severity, on the other hand, identifies the single most hazardous segment along the route.

$$MaxSev = \max_{n \in path} severity(n)$$

Equation 9. Formula for maximum severity score

This measure captures the idea that route safety is constrained by the worst conditions seen using Equation 9, much like a chain is only as strong as its weakest link. Although 90% of a route passes over safe Level 1 flooding, a single Level 4 stretch makes the whole route impassable. MaxSev thus offers a conservative approach safety measure that stops hopeful evaluations depending solely on mean circumstances. Through policy integration, the metric directly affects agent decision-making: when $MaxSev \geq 3$ during route execution, the agent switches from target-seeking mode to shelter-seeking mode, giving current safety precedence over destination arrival. Categorical safety labels based on maximum severity thresholds are offered via route classification. When MaxSev is less than or equal to 2, routes are categorized as Safe; the designation implies that all flooding is either passable (L1) or challenging but manageable with high-clearance vehicles (L2); therefore, the route is advised for broad evacuation. When $MaxSev = 3$, a caution

classification applies, indicating the path contains difficult flooding that requires pedestrian support or automobile care. Containing chest-deep water where vehicle submersion is probable and there is a high drowning threat for pedestrians, dangerous roads have MaxSev = 3; these routes trigger shelter evacuation procedures and are not advised. Impermeable paths with MaxSev = 4 are instantly rejected; hence, the agent must re-plan or evacuate, as such paths present life-threatening full submersion conditions, necessitating evacuation to the closest accessible shelter. Traditional route safety metrics complement severity analysis with quantitative performance measures. Success rate (SR) is defined as:

$$SR = \frac{\text{Episodes reaching target OR shelter}}{\text{Total episodes}} \times 100\%$$

Equation 10. Formula for success rate

In Equation 10, acknowledging that getting to safety is most important even if the original destination proves unreachable due to flooding, this statistic counts both destination arrivals and shelter evacuations as successful outcomes.

3.3.2 System Prototype Development

3.3.2.1 Software Architecture

The system architecture was designed as a multi-layered structure composed of four primary layers: client-side, server-side, database management, and machine learning integration. This layered configuration ensures modularity, clear separation of responsibilities, and efficient system communication.

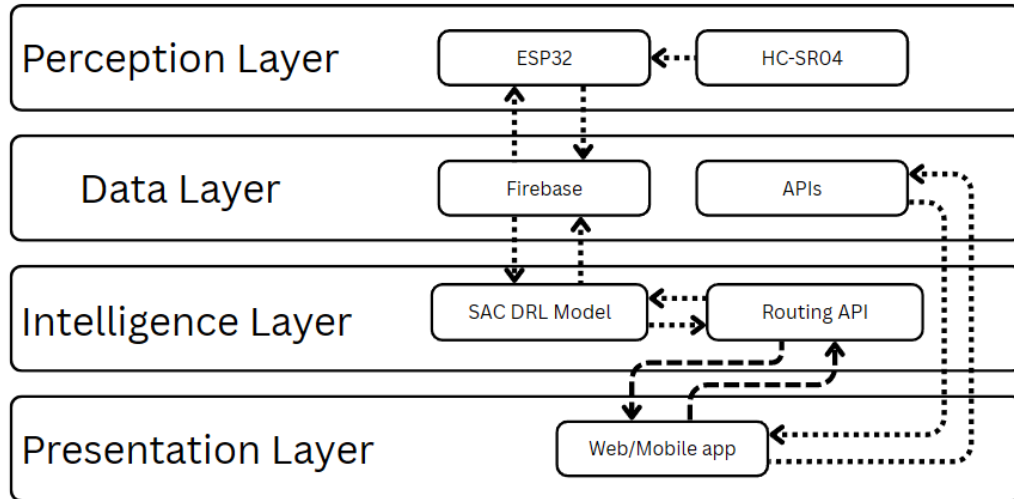


Figure 2. Layered System Architecture Diagram of the Study

Figure 2 illustrates the four-layer architecture of the system: perception layer, data layer, intelligence layer, and presentation layer. As far as the Perception Layer is concerned, it handles the sensing of the environment using ESP32 and HC-SR04 for acquiring the data. The Data Layer handles the communication and storing of the data using Firebase and APIs in the system. The Intelligence Layer is involved in making the decisions using the SAC DRL Model and Routing API to create adaptive responses. Lastly, the Presentation layer offers a user-friendly interface through a web application or mobile application, providing seamless interaction across all components of the system.

3.3.2.2 Client-Side Architecture

The client-side architecture was implemented using a modular JavaScript framework that separates logic into independent functional layers. The Main Application Controller manages module initialization, inter-module communication, and overall application flow. Supporting modules include the API Manager, Map Manager, Data Manager, and User Interface (UI) Manager, each performing specialized roles while maintaining interconnectivity through event-driven operations.

The interface design utilized HTML5, CSS3, and Tailwind CSS for responsive and accessible front-end development. Leaflet.js was used for interactive mapping and spatial data visualization, while Font Awesome provided consistent iconography. This structure ensures a responsive and

dynamic interface capable of rendering live data and ML model outputs efficiently.

3.3.2.3 Server-Side Architecture

The server-side component was implemented using a lightweight and platform-independent configuration. A Python-based HTTP server was employed for static file serving, while a custom CORS proxy server handled cross-origin API requests and facilitated secure communication between the client and third-party services.

An ngrok tunnel was utilized to provide public access during testing and remote evaluation, eliminating the need for permanent server hosting. The server design remains stateless to reduce system overhead and improve maintainability.

3.3.2.4 Database Management Architecture

The system employed a hybrid data storage approach that combined browser-based `localStorage`, in-memory caching, and external API retrieval. `LocalStorage` was used to retain user session data and preferences, while the in-memory cache temporarily stored frequently accessed API responses to minimize redundant network requests.

Data persistence was intentionally limited to short-term caching to maintain lightweight operations. Automatic cleanup mechanisms were implemented to remove outdated or unused data, ensuring consistent performance and efficient memory utilization. This structure enables real-time responsiveness and eliminates the complexity associated with traditional database management systems.

3.3.2.5 Machine Learning Integration Architecture

The system included a forward-compatible design for machine learning integration to support flood prediction and route optimization. RESTful endpoints were defined within the API Manager to facilitate communication between the client-side application and the ML service. A data processing pipeline was established to standardize inputs such as weather, elevation, and traffic information for model compatibility. The architecture incorporated model versioning and A/B testing capabilities, allowing for comparative analysis of

multiple ML models and ensuring traceable performance evaluation. The integration design supports future deployment of deep learning models developed in frameworks such as PyTorch or TensorFlow.

3.3.3 IoT Firmware Architecture

This research employs an integrated suite of hardware, software, and cloud-based instruments to establish a scalable Internet of Things (IoT) framework for real-time water level monitoring and flood detection. The instruments collectively enable continuous environmental data collection, secure cloud synchronization, and dynamic visualization through a web-based interface. The research instruments are classified into two main categories: hardware instruments and digital and data collection instruments.

3.3.3.1 IoT and Firmware Modules

This study utilizes HC-SR04 ultrasonic sensors as the primary instruments for water level monitoring and environmental data acquisition. The HC-SR04 is widely recognized for its affordability, measurement accuracy, and reliability in short-range distance sensing. It operates by emitting ultrasonic pulses and calculating the time interval between transmission and echo reception. This duration is measured in microseconds using the `pulseIn()` function provided by the Arduino framework [1]. The resulting distance is computed based on the speed of sound, achieving an accuracy of approximately ± 3 mm and an effective range between 2 cm and 400 cm, making it appropriate for flood detection and depth measurement applications [2]. The sensor's Trig and Echo configuration allows seamless integration with ESP32 microcontrollers, enabling stable digital input and output operations for precise time-of-flight computations [3]. These sensors will be strategically deployed at designated monitoring locations to gather continuous and objective environmental data, supporting both system calibration and real-world validation.

The ESP32 microcontroller functions as the core of the data acquisition and communication system within the sensor network. It was selected for its dual-core processor, integrated Wi-Fi (IEEE 802.11 b/g/n) capabilities, and extensive General Purpose Input/Output (GPIO) configuration, which allows simultaneous handling of multiple sensor inputs and efficient real-time data transmission [4].

The ESP32 operates on the Arduino platform, utilizing the WiFi.h library for network connectivity and the Firebase ESP Client library for secure HTTPS communication with the Firebase Realtime Database [1]. Each ESP32 node performs data validation and formatting locally before transmitting measurements to the cloud. Each node is assigned a unique identifier, such as sensor_001 or sensor_002, to ensure traceability and scalability within the distributed network. This configuration enables low-latency data transmission, reliable cloud synchronization, and efficient management of multiple monitoring nodes across the study area.

3.3.3.2 Digital and Data Collection Modules

The cloud infrastructure of the monitoring system is built on Firebase Realtime Database (RTDB), a NoSQL cloud database service that provides real-time data synchronization across connected devices [1].

3.3.3.3 Firebase Realtime Database System

The database stores environmental sensor readings as structured JSON objects using hierarchical paths such as sensor/sensor_001/distance, allowing efficient data organization and retrieval [1]. The ESP32 devices communicate with Firebase using HTTPS requests through the REST API, authenticated via a legacy database secret key in accordance with Firebase's REST API documentation [1]. This authentication approach enables secure and consistent data transfer between the distributed sensors and the centralized cloud repository.

During system testing, permissive rules are applied within the Firebase security configuration by setting read and write permissions to true, as documented in Firebase's configuration guidelines [1]. These rules will later be replaced with more restrictive, authentication-based access controls once the system transitions to production. The Firebase RTDB provides high availability, real-time synchronization, and consistent accessibility of environmental data throughout the project lifecycle.

3.3.4 Design Methodology

The development process followed a hybrid methodology combining Rapid Prototyping and Agile Development principles. This approach facilitated rapid iteration, modular feature implementation, and continuous improvement through user feedback and performance evaluation.

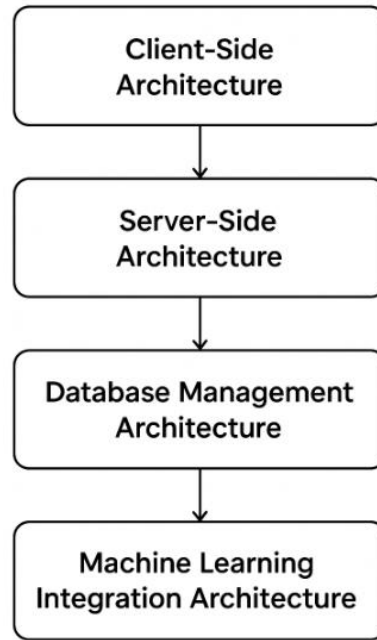


Figure 3. Design Methodology Diagram of the Study

The following figure shows a hybrid design strategy employed in developing the system, using principles of Rapid Prototyping and Agile Development. The process includes iterative design, early visualization, and evolutionary cycles of sprints as its main strategies. Every phase is based on the previous one, using continual testing, feedback from users, and progressive iterations with the objective of guaranteeing systematic functionality, system performance, and user satisfaction.

3.3.4.1 Rapid Prototyping Approach

Rapid Prototyping emphasized early visualization of key functionalities and system responsiveness. Initial iterations focused on core mapping, API integration, and route management components. Successive prototypes incorporate real-time data synchronization, error handling, and performance optimization. Continuous user and peer feedback informed iterative refinements to ensure system accuracy and usability.

3.3.4.2 Agile Development Principles

The agile framework structured the development cycle into two-week sprints, each dedicated to specific milestones such as feature completion, interface refinement, or ML model preparation. Adaptive planning allowed dynamic adjustments based on external API dependencies and real-time data

availability. Regular testing, code reviews, and incremental deployments ensured consistent progress and high software reliability.

3.3.5 Technology Stack

The flood-based rerouting system prototype employs a comprehensive technology stack designed to support both current functionality and future machine learning (ML) model integration. The technology selection prioritizes modularity, scalability, maintainability, and cross-platform compatibility. Each component was chosen to facilitate high-performance computation, efficient data exchange, and seamless interaction between the client interface, backend services, and machine learning endpoints.

3.3.5.1 Programming Languages and Runtime Environments

The system uses JavaScript (ECMAScript 2015+) as the primary client-side programming language. It applies modern ES6+ features such as asynchronous operations using `async/await`, modular imports, and object-oriented programming to handle real-time API interactions, user interface logic, and communication with machine learning services through RESTful endpoints [1].

Python 3.9+ is used for backend development. It provides reliable server-side functions such as HTTP serving, CORS proxy management, and ML endpoint orchestration. Python's library ecosystem, including `requests`, `Flask`, and `NumPy`, enables fast development of data processing pipelines and supports integration with numerical computing frameworks [2].

HTML5 defines the structural and semantic layout of the application, providing accessibility features and geolocation APIs for enhanced user experience. CSS3 defines the presentation layer, using responsive design methods such as `Flexbox`, `Grid`, and media queries to ensure compatibility across various devices and resolutions [3].

3.3.5.2 Frontend Frameworks and Libraries

The frontend architecture uses several frameworks and libraries to improve performance and design consistency. `Tailwind CSS v3.x`, a utility-first CSS framework, is applied for atomic and responsive styling, promoting uniform design while optimizing load times through a purge mechanism [4].

Leaflet.js v1.9.x supports interactive maps and geospatial visualization, managing map layers, markers, and overlays for real-time data display []. The Leaflet Awesome Markers plugin extends Leaflet by integrating Font Awesome v6.x, providing scalable and visually clear icons for geographic representation. This combination improves interface clarity and usability.

3.3.5.3 Backend Infrastructure and Services

The backend supports data exchange, proxy communication, and secure network interaction using lightweight tools. The Python HTTP Server module is used for local hosting and static file serving during development. A Custom CORS Proxy Server, implemented with the `http.server` and `requests` libraries, manages cross-origin restrictions and maintains secure communication with external APIs.

The ngrok tunneling service allows external access to local servers through secure channels. This setup makes it possible to test and demonstrate the system remotely in real time [].

3.3.5.4 External API Integrations and Data Sources

The system depends on several external APIs for real-time environmental and geographic data. The OpenWeatherMap API v2.5 provides meteorological information including weather forecasts, temperature, and humidity data []. The Google Maps Elevation API offers precise elevation data that supports flood-risk assessment and terrain-based route planning [].

OpenStreetMap (OSM) supplies open-source geographic data such as road networks and base map tiles. The TomTom Traffic API provides live traffic flow and incident data that improves route accuracy during emergencies []. The Firebase Realtime Database supports synchronization of live environmental sensor inputs with the flood prediction model, ensuring real-time awareness and data consistency [].

3.3.5.5 Development Tools and Environment

The development setup focuses on flexibility and productivity. Visual Studio Code (VS Code) functions as the main integrated development environment with built-in debugging, Git integration, and live server support. Git

manages source control, allowing structured collaboration with branching and commit tracking [1].

Windows Batch Scripts automate repetitive development tasks such as server startup and synchronization. Browser Developer Tools, including Chrome DevTools and Firefox Developer Tools, assist in debugging, monitoring network requests, and improving runtime performance.

3.3.5.6 Machine Learning Integration Infrastructure

The machine learning infrastructure is designed for consistency, communication, and performance evaluation. A RESTful API architecture maintains standardized communication between the client modules and backend ML services [1].

A Custom JavaScript Data Processing Pipeline cleans and normalizes environmental data before inference. JSON is used as the standard format for data transfer and configuration. A Model Versioning System allows tracking of multiple ML models for regression testing and comparison. An A/B Testing Framework provides a controlled method to evaluate and compare different ML models under the same operating conditions.

3.3.5.7 Deployment and Distribution Architecture

Deployment emphasizes accessibility, portability, and reproducibility. The Python HTTP Server supports local deployment and static file hosting [1]. The ngrok service provides secure remote access to live environments during testing. Multi-platform compatibility ensures that the system can operate on Windows, macOS, and Linux through its browser-based design and Python runtime portability.

Static file hosting allows deployment to cloud or shared web services without depending on an active backend, simplifying maintenance and scalability [1].

3.3.5.8 Security and Performance Considerations

Security and performance optimization are built into the system architecture. The CORS Proxy Security mechanism verifies communication between client and server components and maintains encryption for data integrity. API Key Management stores credentials in environment-based configuration files to prevent unauthorized access.

Client-side caching with time-to-live parameters reduces redundant API requests and improves response speed. Asynchronous processing with Promises and `async/await` provides smoother user interactions. Finally, Tailwind CSS optimization through purging and minification minimizes the resource footprint and improves load performance [1].

3.3.6 Module Development

The Flood-Based Rerouting System was designed and implemented using a modular development approach to ensure maintainability, scalability, and adaptability. Each module is responsible for a distinct function within the system, but all operate cohesively through event-driven communication and shared data interfaces.

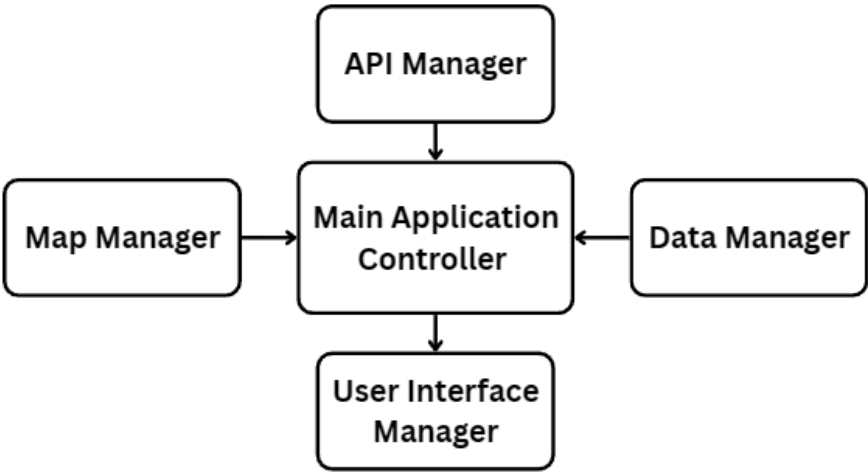


Figure 4. Module Interaction Diagram of the Study

Figure 4 shows how the core modules of the system interact, featuring the Client Layer, Application Layer, and Database Layer. It demonstrates the process of handling user requests starting at the client interface. These requests pass through application logic for validation and processing before reaching the database for data storage or retrieval.

3.3.6.1 Main Application Controller

The Main Application Controller manages the lifecycle of all system modules. It initializes components, coordinates inter-module communication, and handles event propagation. This module oversees geolocation services, route computation, environmental data updates, and ML model integration. Its asynchronous design ensures non-blocking operation and real-time responsiveness to dynamic environmental data changes.

3.3.6.2 API Manager Module

The API Manager Module is responsible for all external data communication and management. It handles RESTful requests to APIs such as OpenWeatherMap, Google Maps Elevation, TomTom Traffic, and Firebase. It also processes authentication, rate limiting, and error handling. This module implements caching and retry mechanisms to ensure data reliability, while managing preprocessing for ML input data to ensure compatibility with the model's expected format.

3.3.6.3 Map Manager Module

The Map Manager Module handles geospatial visualization and interactive mapping. Using Leaflet.js, it renders map layers, routes, and flood-risk overlays derived from ML predictions. The module supports dynamic updates to reflect real-time changes in flood or traffic conditions. It provides layer control for satellite and terrain views, integrates marker clustering for clarity, and visualizes predictive data through color-coded overlays and route adjustments.

3.3.6.4 Data Manager Module

The Data Manager Module manages local data caching, synchronization, and persistence. It employs localStorage for user-specific settings and in-memory caching for high-frequency API responses. It enforces data integrity through validation and sanitization processes. The module also acts as a bridge between the API Manager and ML Integration Module, ensuring structured data transfer and temporary dataset retention during computation.

3.3.6.5 User Interface Manager Module

The User Interface Manager Module controls all aspects of the client-side interface. It manages dynamic components such as sidebars, modals, and tooltips, and supports user input handling. Accessibility and responsive design are core considerations, ensuring usability across devices. It also provides real-time

visualization of flood predictions and route adjustments from ML outputs, updating indicators, alerts, and notifications dynamically.

3.3.6.6 Machine Learning Integration Module

The Machine Learning Integration Module facilitates data exchange and inference operations between the client and the ML model. It preprocesses input data, formats requests for RESTful endpoints, and handles responses for visualization and decision-making. The module supports multiple ML model versions, enabling model comparison through built-in A/B testing. Additionally, it includes performance tracking tools to monitor prediction latency and accuracy during real-time operation.

3.4 THEORETICAL AND CONCEPTUAL FRAMEWORK

3.4.1 Theoretical Framework

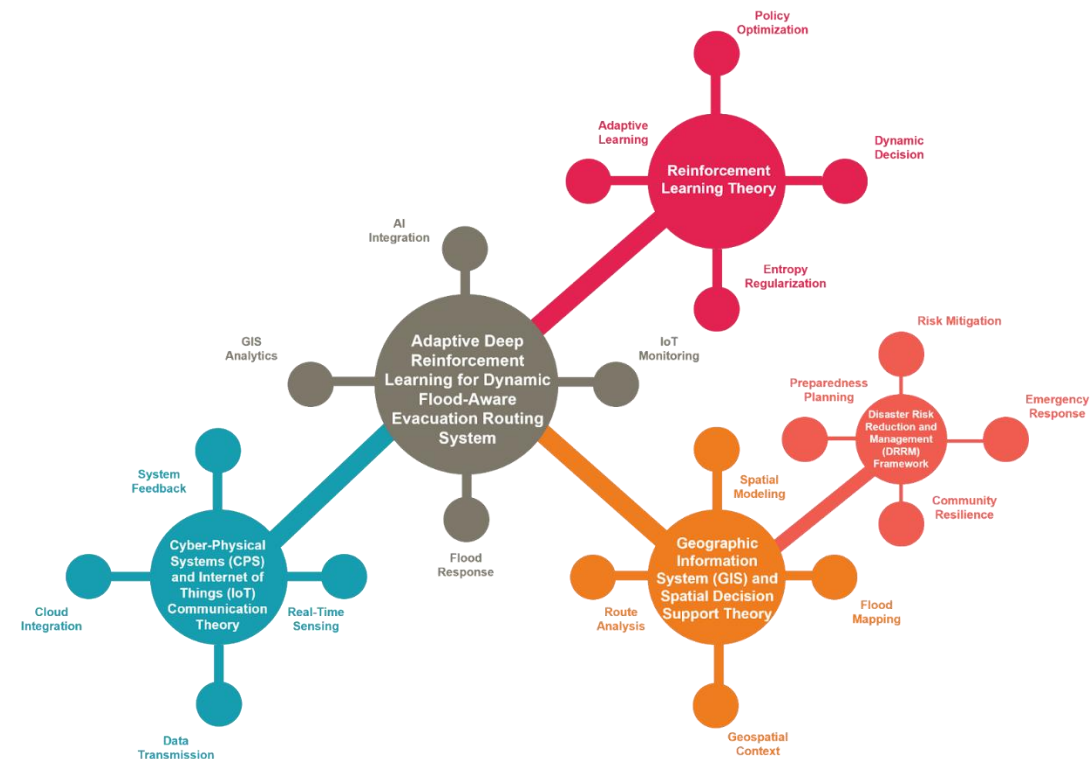


Figure 5. Theoretical Framework of the Study

Figure 5 presents the theoretical foundations that support the development of the *Adaptive Deep Reinforcement Learning for Dynamic Flood-Aware Evacuation Routing System*. The diagram illustrates the integration of four major theories: Reinforcement Learning Theory, Geographic Information System and Spatial Decision Support Theory, Disaster Risk Reduction and Management (DRRM) Framework, and

Cyber-Physical Systems (CPS) and Internet of Things (IoT) Communication Theory. Each theory contributes distinct principles to the study: Reinforcement Learning Theory provides the foundation for adaptive decision-making; GIS and Spatial Decision Support Theory guide spatial modeling and route analysis; the DRRM Framework grounds the system in disaster preparedness, mitigation, and response; while CPS and IoT Communication Theory ensure real-time sensing, cloud integration, and continuous feedback. Their interconnection forms the scientific basis for developing an intelligent, real-time, and flood-aware evacuation routing system tailored for urban flood management in Manila City.

The study is grounded in four major theoretical pillars, namely: Reinforcement Learning Theory, Geographic Information System and Spatial Decision Support Theory, Disaster Risk Reduction and Management (DRRM) Framework, and Cyber-Physical Systems (CPS) and Internet of Things (IoT) Communication Theory. These theoretical bases collectively form the scientific structure that explains how intelligent computation, spatial analysis, and real-time environmental monitoring can be combined to produce an adaptive and responsive flood evacuation system.

Reinforcement Learning Theory provides the computational foundation of the study. It originates from behavioral psychology and machine learning, emphasizing how an agent learns optimal behavior through interaction with its environment via trial and error. The agent receives rewards for desirable actions and penalties for undesirable ones and refines its actions to maximize long-term cumulative reward, as detailed in [1]. In this study, the flood evacuation process is modeled as a Markov Decision Process, where the agent learns to navigate a dynamic, flood-prone environment by selecting routes that minimize exposure to risk while maximizing safety and efficiency. The Soft Actor-Critic algorithm, an entropy-regularized reinforcement learning approach, promotes both exploration and adaptability (discussed in literature on deep RL). This theory supports the use of adaptive learning to manage uncertainty in real-world flood conditions, ensuring that the system continuously improves through environmental feedback.

Geographic Information System and Spatial Decision Support Theory establish the spatial and analytical foundation of this study. This theory emphasizes the integration of spatial data for informed decision-making when geographic relationships and spatial constraints are present. The integration of spatial decision support technologies within GIS enables data-driven modeling, as explored in spatial decision support literature. In this study, GIS constructs a spatially explicit simulation

environment combining road networks, elevation data, and flood depth measurements into a flood-augmented graph. Each route or edge within this graph is weighted by parameters representing distance, flood intensity, and hazard exposure. This spatial representation allows the agent to perceive real-world constraints and make geographically informed decisions. The theory ensures that artificial intelligence operates within a realistic, data-based geospatial framework, enabling accurate route optimization in flood-prone urban environments.

The Disaster Risk Reduction and Management (DRRM) Framework provides the humanitarian and policy dimension of the study. This framework promotes a structured approach to minimizing disaster risk through stages of preparedness, mitigation, response, and recovery, consistent with the Sendai Framework for Disaster Risk Reduction [1]. This study aligns with these principles by developing an adaptive, technology-enabled system that enhances situational awareness and decision-making during flood emergencies. Through real-time data collection and intelligent routing, the proposed system contributes to improved disaster preparedness and emergency response, ensuring that evacuation routes are safe, accessible, and context-aware. This theoretical alignment underscores the relevance of integrating scientific innovation with disaster management practices to promote community resilience and reduce risk.

Cyber-Physical Systems and IoT Communication Theory provide the technological and operational foundation of the study. These theories explain how interconnected physical devices and computational models operate collaboratively to enable intelligent, real-time decision-making. The concept of CPS discusses the coupling of computation, networking, and physical systems [2]. In this study, IoT-enabled sensors such as HC-SR04 ultrasonic sensors connected via ESP32 microcontrollers serve as the physical data collection layer, transmitting real-time flood depth information to a cloud database (Firebase). This integration bridges physical sensing with digital decision-making, enabling continuous feedback and dynamic response to environmental changes. The theoretical foundation validates the system's use of CPS and IoT principles to maintain responsiveness, synchronization, and adaptability under rapidly changing flood conditions.

The synthesis of these theoretical perspectives forms the complete intellectual foundation of the study. Reinforcement Learning Theory ensures that the system learns adaptively through environmental interaction. GIS and Spatial Decision Support Theory ensure that the system's decisions are spatially coherent and grounded in

spatial data. The DRRM Framework provides societal relevance and humanitarian rationale, while CPS and IoT Communication Theory undergo real-time sensing and feedback mechanisms. Together, these theories form a coherent and integrated framework that justifies the design, operation, and significance of an adaptive, flood-aware evacuation routing system in urban flood-prone areas.

This figure establishes the conceptual and scientific foundation of the study by integrating computational intelligence, spatial analytics, disaster management principles, and real-time sensing technologies. These theories collectively explain how the proposed system enhances evacuation safety, operational reliability, and adaptive decision-making during flood emergencies. The succeeding diagram illustrates the interconnections among these theories and their relationship to the study’s central components, serving as a scientific mapping diagram of the theoretical framework.

3.4.2 Conceptual Framework

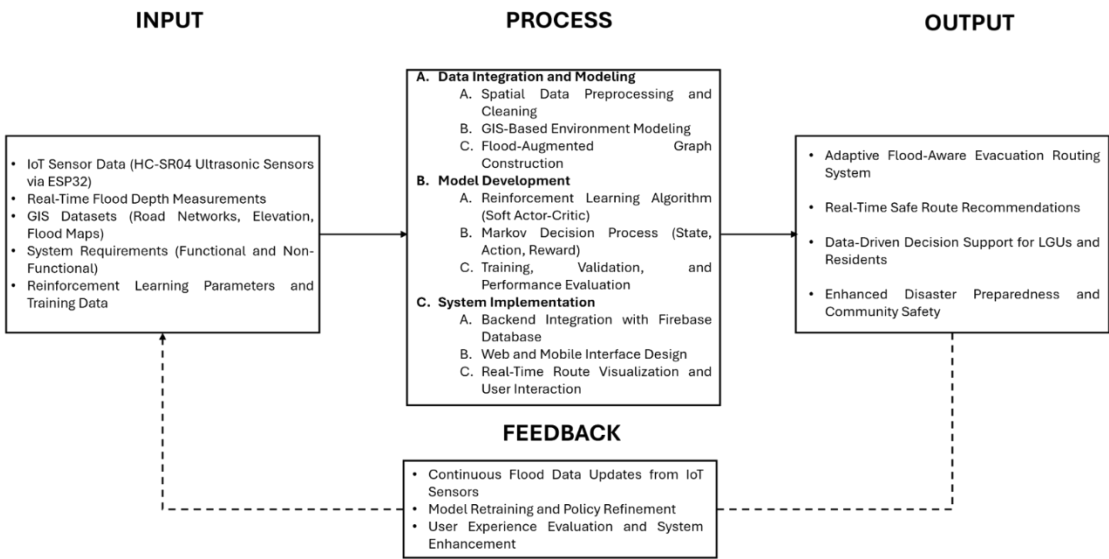


Figure 6. Conceptual Framework of the Study

Figure 6 illustrates the Input–Process–Output (IPO) model with Feedback representing the conceptual framework of the study. The Input stage consists of IoT sensor data, GIS datasets, reinforcement learning parameters, and system requirements. The Process stage includes three core components: data integration and modeling, model development using the Soft Actor-Critic algorithm, and system implementation integrating backend, database, and interface layers. The Output produces an adaptive flood-aware evacuation routing system capable of generating real-time safe routes and

supporting local decision-makers and residents during flood events. The Feedback mechanism ensures continuous improvement through updated flood data, reinforcement learning retraining, and user evaluation, thereby maintaining system accuracy, reliability, and responsiveness in real-world conditions. The Feedback mechanism ensures continuous improvement through updated flood data, reinforcement learning retraining, and user evaluation, thereby maintaining system accuracy, reliability, and responsiveness in real-world conditions.

In the first stage, Data Acquisition, real-time environmental and flood depth data are collected with IoT sensors such as HC-SR04 ultrasonic modules interfaced with ESP32 microcontrollers. These sensors transmit data to a cloud database (e.g. Firebase). This stage embodies cyber-physical systems and IoT communication theory, where physical sensing devices feed data into computational systems in real time [1].

In the second stage, Data Integration and Processing, spatial datasets (road networks, elevation models, flood maps) are combined to create a flood-augmented graph model of the urban environment. This aligns with the principles of GIS and spatial decision support, where integrated spatial data support analytical modeling [2].

The third stage, Decision Intelligence, is the computational core. Guided by reinforcement learning theory, the Soft Actor-Critic algorithm is used to train an agent that navigates the GIS-based environment, learning optimal evacuation policies under dynamic flood conditions [3, 4].

In the fourth stage, User Interaction and Response, computational outputs are converted into actionable guidance via web and mobile interfaces. The system presents real-time evacuation routes, safety alerts, and situational updates to users and decision-makers. This stage ties into the DRRM framework, emphasizing that technical outputs must support public safety, preparedness, and responsive decision making [5].

Lastly, the framework ends with the Expected Outputs, which include an adaptive routing system capable of identifying safe evacuation paths dynamically, minimizing evacuation time, and promoting resilience in flood-prone communities. The system's outputs also serve local government units as analytical tools for proactive disaster management.

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APPENDICES

APPENDIX A.

LETTERS AND

DOCUMENTATIONS

APPENDIX B. TRANSCRIPT OF INTERVIEWS AND PHOTO DOCUMENTATION

APPENDIX C.

PANEL'S LIST OF RECOMMENDATIONS

